

§ 2.2 换元法

$$\int e^x dx = e^x + C$$

$$\begin{aligned} \textcircled{13} \quad \int e^{2x} dx &= \frac{1}{2} \int e^{2x} \cdot 2 dx = \frac{1}{2} \int e^{2x} d(2x) \\ &= \frac{1}{2} \int e^u du = \frac{1}{2} e^u + C \\ &= \frac{1}{2} e^{2x} + C. \end{aligned}$$

$u=2x$
↓

$$\textcircled{14} \quad \int \frac{1}{1+x} dx \quad \int \frac{1}{x} dv = \ln|x| + C.$$

↓

$$d(1+x) = dx$$

$$= \int \frac{1}{1+x} d(1+x) \quad \begin{array}{c} u=1+x \\ \hline \downarrow \end{array}$$

$$= \int \frac{1}{u} du = \ln|u| + C = \ln|1+x| + C.$$

$$f'(u) = f(u), \quad u = \varphi(x).$$

$$dF(\varphi(x)) = dF(u) = f(u) du = f(\varphi(x)) \varphi'(x) dx$$

$$\int f(\varphi(x)) \varphi'(x) dx = \int f(u) du$$

$$\int f(\varphi(x)) d\varphi(x) = \int f(u) du$$

(15) $\int \frac{\ln x}{x} dx$ $\frac{1}{x} dx = d \ln x$

$= \int \ln x d \ln x$ $\xrightarrow{u = \ln x}$ $= \int u du = \frac{1}{2} u^2 + C = \frac{1}{2} (\ln x)^2 + C.$

(16) $\int \underline{x e^{x^2}} dx$ $\textcircled{x^2}$ $d x^2 = 2x dx$

$= \frac{1}{2} \int e^{x^2} \cdot \underline{2x dx} = \frac{1}{2} \int e^{x^2} d(x^2)$

$\xrightarrow{\textcircled{u = x^2}}$

$= \frac{1}{2} \int e^u du = \frac{1}{2} e^u + C$

$= \frac{1}{2} e^{x^2} + C.$

$$\begin{aligned}
 (17) \quad \int \underline{\sin^3 x} \underline{\cos x} \, dx &= \int \sin^3 x \cdot \underline{(\sin x)'} \, dx \\
 &= \int \sin^3 x \cdot d \sin x \quad u = \sin x \\
 &= \underline{\int u^3 \, du} = \frac{1}{4} u^4 + C \quad \downarrow \\
 &= \frac{1}{4} \sin^4 x + C.
 \end{aligned}$$

$$\begin{aligned}
 (18) \quad \int \sin^3 x \, dx &= \int \underline{\sin^2 x} \cdot \underline{\sin x} \, dx \\
 &= -\int \underline{\sin^2 x} \cdot d \underline{\cos x} = -\int \underline{(1 - \cos^2 x)} \, d \cos x \\
 &= -\int d \cos x + \int \cos^2 x \, d \cos x \\
 &= -\cos x + \frac{1}{3} \cos^3 x + C.
 \end{aligned}$$

$$(19) \quad \int \cos^2 x \, dx = \frac{1}{2} \int (1 + \cos 2x) \, dx \quad \cos 2x = 2\cos^2 x - 1$$

$$= \frac{1}{2} x + \frac{1}{2} \int \cos 2x \, dx$$

$$= \frac{1}{2} x + \frac{1}{4} \int \cos x \, d 2x = \frac{1}{2} x + \frac{1}{4} \sin 2x + C.$$

$$(20) \cdot \int \sin^2 x \cos^5 x \, dx$$

$$= \int \sin^2 x \cdot \cos^4 x \cos x \, dx$$

$$= \int \sin^2 x \cdot \cos^4 x \, d \sin x$$

$$= \int \sin^2 x (1 - \sin^2 x)^2 \, d \sin x$$

$$= \int \sin^2 x - 2 \sin^4 x + \sin^6 x \, d \sin x$$

$$= \frac{1}{3} \sin^3 x - \frac{2}{5} \sin^5 x + \frac{1}{7} \sin^7 x + C$$

$$(21) \int \tan^n x \, dx = \int \frac{\sin x}{\cos x} \, dx$$

$$\begin{aligned}
 (21) \quad \int \frac{1}{\cos x} dx &= \int \frac{1}{u} du = \ln|u| \\
 &= - \int \frac{(\cos x)'}{\cos x} dx = - \int \frac{1}{\cos x} d(\cos x) \\
 &= -\ln|\cos x| + C \\
 &= \ln\left|\frac{1}{\cos x}\right| + C \\
 &= \ln|\sec x| + C.
 \end{aligned}$$

$$(22) \cdot \int \frac{1}{1+e^x} dx$$

①.

$$\begin{aligned}
 d(1+e^x) &= e^x dx \\
 &= d e^x
 \end{aligned}$$

$$\begin{aligned}
 \int \frac{1+e^x - e^x}{1+e^x} dx &= \int 1 dx - \int \frac{e^x}{1+e^x} dx \\
 &= x - \int \frac{1}{1+e^x} d(1+e^x),
 \end{aligned}$$

$$\int \frac{1}{u} du$$

$$= x - \ln(1+e^x) + C.$$

$$\textcircled{23}. \int \frac{e^{-x}}{e^{-x}+1} dx$$

$$d(e^{-x}+1) = -e^{-x} dx$$

$$= - \int \frac{-e^{-x} dx}{e^{-x}+1} = - \int \frac{1}{e^{-x}+1} d(e^{-x}+1)$$

$$= -\ln(e^{-x}+1) + C.$$

$\textcircled{24}. \quad \textcircled{4.2.20}$
 $g(x) \neq 0,$

$$\int \frac{g'(x)}{g(x)} dx = \int \frac{1}{g(x)} dg(x) = \ln|g(x)| + C$$

$$\textcircled{25}. \int \frac{dx}{ax+b} \quad (a \neq 0)$$

$$d(ax+b) = a dx$$

$$) \quad ax+b$$

$$= \frac{1}{a} \int \frac{a \, dx}{ax+b} = \frac{1}{a} \int \frac{1}{ax+b} \, d(ax+b)$$

$$= \frac{1}{a} \ln |ax+b| + C.$$

$$(26) \quad \int \frac{x}{1+x^2} \, dx \quad d(1+x^2) = 2x \, dx$$

$$= \frac{1}{2} \int \frac{2x}{1+x^2} \, dx = \frac{1}{2} \int \frac{(1+x^2)'}{(1+x^2)} \, dx$$

$$= \frac{1}{2} \ln(1+x^2) + C.$$

$$(27). \quad \int \arctan x \, dx \quad \frac{\int x \arctan x \, dx}{u \, dv}$$

$$= x \arctan x - \int x \, d(\arctan x)$$

$$= x \arctan x - \int \frac{x}{1+x^2} dx$$

$$= x \arctan x - \frac{1}{2} \ln(1+x^2) + C.$$

$$(28) \int \frac{1}{x(1+2\ln x)} dx$$

$$d(1+2\ln x) = \frac{2}{x} dx$$

$$\begin{aligned} &= \frac{1}{2} \int \frac{\frac{2}{x}}{1+2\ln x} dx = \frac{1}{2} \int \frac{(1+2\ln x)'}{1+2\ln x} dx \\ &= \frac{1}{2} \ln |1+2\ln x| + C. \end{aligned}$$

§ 换元法

设 $f(u)$ 有原函数 $F(u)$ $(F'(u) = f(u))$

$u = \varphi(x)$ 可导.

$$\begin{aligned} dF(\varphi(x)) &= F'(\varphi(x)) \varphi'(x) dx = f(\varphi(x)) \varphi'(x) dx \\ &= dF(u) = F'(u) du = f(u) du. \end{aligned}$$

$$f(\varphi(x)) \varphi'(x) dx = f(u) du$$

$$\int f(\varphi(x)) \varphi'(x) dx = \boxed{\int f(u) du} \quad \text{①.}$$

$$\int f(\varphi(x)) d\varphi(x) = \int f(u) du \quad \text{②.}$$

§ 第一换元法 (凑微分法)

$$\int \underline{\underline{u'(x)}} dx$$

$$\begin{array}{c} \wedge \\ f(\varphi(x)) \quad \varphi'(x) \end{array}$$

例 ①. $\int \frac{dx}{a^2 + x^2} \quad (a \neq 0).$

~~若~~ $a=1 \quad \int \frac{1}{1+x^2} dx = \arctan x + C.$

$$\int \frac{1}{a^2 + x^2} dx = \int \frac{1}{\textcircled{a^2} [1 + (\frac{x}{a})^2]} \underline{\underline{dx}}$$

$$d\frac{x}{a} = \frac{1}{a} dx$$

$$= \frac{1}{a} \int \frac{1}{1 + (\frac{x}{a})^2} \cdot \frac{1}{a} dx$$

$$= \frac{1}{a} \int \frac{1}{1 + (\frac{x}{a})^2} d(\frac{x}{a})$$

$$= \frac{1}{a} \arctan\left(\frac{x}{a}\right) + C.$$

$$\textcircled{2}. \int \frac{1}{\sqrt{a^2 - x^2}} dx \quad (a > 0).$$

$$\xrightarrow{\text{if } a=1} \int \frac{1}{\sqrt{1-x^2}} dx = \operatorname{arcsinh} x + C.$$

$$= \int \frac{1}{\sqrt{1 - \left(\frac{x}{a}\right)^2}} \frac{1}{a} dx = \int \frac{1}{\sqrt{1 - \left(\frac{x}{a}\right)^2}} d\left(\frac{x}{a}\right) \quad \underline{u = \frac{x}{a}}$$

$$= \operatorname{arcsinh}\left(\frac{x}{a}\right) + C.$$

$$\textcircled{3} \int \frac{1}{a^2 - x^2} dx \quad (a \neq 0)$$

$$= \int \frac{1}{(a-x)(a+x)} dx$$

$$= \frac{1}{2} \int \left(\frac{1}{a-x} - \frac{1}{a+x} \right) dx$$

$$\frac{1}{2a} \int \left[\frac{1}{a-x} + \frac{1}{a+x} \right] dx$$

$$= \frac{1}{2a} \int \frac{1}{a+x} dx + \int \frac{1}{a-x} dx$$

$$(a+x)' = 1$$

$$(a-x)' = -1$$

$$= \frac{1}{2a} \int \frac{(a+x)'}{a+x} dx - \frac{1}{2a} \int \frac{(a-x)'}{a-x} dx = \frac{1}{2a} \int \frac{1}{a+x} d(a+x) - \frac{1}{2a} \int \frac{1}{a-x} d(a-x)$$

$$= \frac{1}{2a} \left[\ln |a+x| - \ln |a-x| \right] + C$$

$$= \frac{1}{2a} \ln \left| \frac{a+x}{a-x} \right| + C$$

$$\textcircled{4} \int \frac{1}{a^2 \sin^2 x + b^2 \cos^2 x} dx \quad (ab \neq 0)$$

$$(\tan x)' = \frac{1}{\cos^2 x}$$

$$d \tan x = \frac{1}{\cos^2 x} dx$$

$$= \int \frac{1}{\cos^2 x \left(\frac{a^2 \sin^2 x}{\cos^2 x} + b^2 \right)} dx = \int \left(\frac{1}{\cos^2 x} \right) \cdot \frac{1}{a^2 \tan^2 x + b^2} (dx)$$

$$= \int \frac{1}{a^2 \tan^2 x + b^2} d(\tan x)$$

$$\begin{aligned}
 &= \int \frac{1}{a^2} \frac{1}{\tan^2 x + \left(\frac{b}{a}\right)^2} d \tan x \\
 &= \left(\frac{1}{a^2}\right) \int \frac{1}{\tan^2 x + \left(\frac{b}{a}\right)^2} d \tan x \\
 &= \frac{1}{ab} \cdot \arctan \left(\frac{a}{b} \tan x \right) + C.
 \end{aligned}$$

⑤ $\int \sec x dx$ 及 $\int \csc x dx$

$$\underline{d \sinh x = \cosh x dx}$$

解: ①. $\int \sec x dx = \int \frac{1}{\cos x} dx = \int \frac{\frac{\cos x}{\cos^2 x} dx}{\cos^2 x} = \int \frac{1}{\cos^2 x} d \sinh x$

$$= \int \frac{1}{1 - \sinh^2 x} d \sinh x \quad \int \frac{1}{a^2 - b^2} dx$$

$$= \frac{1}{2} \ln \left| \frac{1 + \sinh x}{1 - \sinh x} \right| + C$$

$$\int \csc x dx = \frac{1}{2} \ln \left| \frac{1 - \cos x}{1 + \cos x} \right| + C.$$

解: ②. $\int \csc x \, dx = \int \frac{1}{\sin x} \, dx = \int \frac{1}{2 \sin \frac{x}{2} \cos \frac{x}{2}} \, dx$

$$\left(\tan \frac{x}{2} \right)' = \frac{1}{2} \cdot \frac{1}{\cos^2 \frac{x}{2}} = \int \frac{1}{\frac{\sin \frac{x}{2}}{\cos^2 \frac{x}{2}} \cdot \underbrace{2 \cos^2 \frac{x}{2}}_{dx}} \, dx$$

$$= \int \frac{1}{\tan \frac{x}{2}} \, d\left(\tan \frac{x}{2}\right)$$

$$\int \frac{1}{u} \, du$$

$$= \ln \left| \tan \frac{x}{2} \right| + C.$$

$$\tan \frac{x}{2} = \frac{\sin \frac{x}{2}}{\cos \frac{x}{2}}$$

$$= \frac{2 \sin \frac{x}{2} \sin \frac{x}{2}}{2 \cos \frac{x}{2} \sin \frac{x}{2}} = \frac{1 - \cos x}{\sin x}$$

$$= \frac{1}{\sin x} - \frac{\cos x}{\sin x} = \csc x - \cot x$$

$$= \frac{1}{\sin x} - \frac{\cos x}{\sin x} = \csc x - \cot x$$

$$= \ln \left| \csc x - \cot x \right| + C.$$

$$\frac{\csc x - \cot x}{1}$$

$$(\csc x + \cot x) = \frac{\csc^2 x - \cot^2 x}{\csc x + \cot x} = \frac{1}{\csc x + \cot x}$$

$$\begin{aligned}
 &= \ln \left| \frac{1}{\csc x + \cot x} \right| + C \\
 &= -\ln |\csc x + \cot x| + C.
 \end{aligned}$$

$$\begin{aligned}
 \tan \frac{x}{2} &= \frac{1 - \cos x}{\sin x} = \frac{1 - \cos x}{\sqrt{1 - \cos^2 x}} = \sqrt{\frac{(1 - \cos x)^2}{1 - \cos^2 x}} \\
 &= \sqrt{\frac{1 + \cos x}{1 - \cos x}}
 \end{aligned}$$

解③. $\int \sec x \, dx$

$$\frac{\sec x}{1}$$

$$(\sec x)' = \underline{\underline{\sec x \tan x}}$$

$$(\tan x)' = \underline{\underline{\sec^2 x}}$$

$$\underline{\underline{(\sec x + \tan x)' = \sec x (\tan x + \sec x)}}$$

$$\int \frac{\sec x (\sec x + \tan x)}{\sec x + \tan x} \, dx$$

$$\int 1 \, d(\sec x + \tan x)$$

$$= \int \frac{1}{\sec x + \tan x} dx$$

$$= \ln |\sec x + \tan x| + C.$$

$$\textcircled{6} \quad \int \sec^3 x \, dx$$

$$= \int \sec x \cdot \sec^2 x \, dx$$

$$= \int \sec x \, d \tan x$$

$$= \sec x \tan x - \int \tan x \, d \sec x$$

$$= \sec x \tan x - \int \tan x \tan x \sec x \, dx$$

$$= \dots - \int \tan^2 x \sec x \, dx$$

$$= \sec x \tan x - \int (\sec^2 x - 1) \sec x \, dx$$

$$= \sec x \tan x - \int \sec^3 x \, dx + \int \sec x \, dx$$

$$\int \sec^3 x \, dx = \frac{1}{2} \sec x \tan x + \frac{1}{2} \int \sec x \, dx$$

$$\sinh^5 x = \sinh^4 x \underbrace{\sinh x \, dx}_{-d \cos x}$$

$$\left\{ \begin{array}{l} \int (\sinh^2 x)^2 (1 - \cosh^2 x)^2 d(-\cosh x) \\ \sinh^2 x = \cosh^2 x - 1 \end{array} \right.$$

$$\sec^2 x = 1 + \tan^2 x$$

$$= \frac{1}{2} \sec x \tan x + \frac{1}{2} \ln |\sec x + \tan x| + C.$$

$$\int \sec^{2n+1} x \, dx \rightarrow T + \int \sec^{2n-1} x \, dx$$

$$\begin{aligned} \int \sec^{2n+2} x \, dx &= \int \sec^{2n} x \cdot \sec^2 x \, dx \\ &= \int (1 + \tan^2 x)^n \, d \tan x \end{aligned}$$

$$\textcircled{7}. \int \frac{1}{1 + \cos x} \, dx$$

$$= \int \frac{1 - \cos x}{1 - \cos^2 x} \, dx = \int \frac{1 - \cos x}{\sin^2 x} \, dx$$

$$= \int \frac{1}{\sin^2 x} \, dx - \int \frac{1}{\sin^2 x} \, d \sin x$$

$$= -\cot x + \frac{1}{\sin x} + C.$$

$$\textcircled{1} \text{式} \quad \int f(\varphi(x)) \varphi'(x) dx = \int f(u) du$$

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$$\underline{\int f(u) du} = \boxed{\int f(\varphi(x)) \varphi'(x) dx} \quad \textcircled{2} \text{式}$$

$\boxed{u = \varphi(x)} \quad \underline{x = \varphi^{-1}(u)}$

证 4.2.2. 设 $f(x)$ 有原函数, $x = \varphi(t)$ 可导, $\varphi'(t) \neq 0$, 且 $\varphi(t)$ 有连续的反函数 $t = \varphi^{-1}(x)$.
 且有 $\underline{\int f(\varphi(t)) \varphi'(t) dt = F(t) + C} \quad \textcircled{4}$.

那么

$$\underline{\int f(x) dx} = \underline{F(\varphi^{-1}(x)) + C} \quad \textcircled{5}$$

证明:

$$\begin{aligned} & \frac{d}{dx} [F(\varphi^{-1}(x)) + C] \\ &= F'(\varphi^{-1}(x)) [\varphi^{-1}]'(x) \\ &= f(\varphi(t)) \cancel{\varphi'(t)} \frac{1}{\cancel{\varphi'(t)}} = f(\varphi(t)) = f(x) \end{aligned}$$

$$\textcircled{1} \int \sqrt{a^2 - x^2} \, dx \quad (a > 0).$$

$$\sin^2 t + \cos^2 t = 1.$$

$$\text{Let } x = a \sin t \quad \left(|t| \leq \frac{\pi}{2} \right).$$

$$\sqrt{a^2 - x^2} = \sqrt{a^2 - a^2 \sin^2 t} = a \cos t.$$

$$dx = a \cos t \, dt.$$

$$\int \sqrt{a^2 - x^2} \, dx = \int a \cos t \cdot a \cos t \, dt$$

$$= a^2 \int \cos^2 t \, dt$$

$$= \frac{a^2}{2} \int (1 + \cos 2t) \, dt$$

$$= \frac{a^2}{2} \left(t + \underline{\underline{\frac{1}{2} \sin 2t}} \right) + C$$

$$= \frac{a^2}{2} (t + \sin t \cos t) + C.$$

$$t = \arcsin \frac{x}{a}$$

$$\sin t \cos t = \sin t \sqrt{1 - \sin^2 t} = \underline{\sin \left(\arcsin \frac{x}{a} \right) \cdot \sqrt{1 - \sin^2 \left(\arcsin \frac{x}{a} \right)}}$$

$$= \frac{x}{a} \cdot \sqrt{1 - \left(\frac{x}{a}\right)^2}$$

$$= \frac{1}{a^2} x \sqrt{a^2 - x^2}$$

$$= \frac{a^2}{2} \operatorname{arcsinh} \frac{x}{a} + \frac{1}{2} x \sqrt{a^2 - x^2} + C.$$

$$\textcircled{2} \int \frac{1}{\sqrt{a^2 + x^2}} dx \quad (a > 0)$$

$$1 + \tan^2 x = \sec^2 x = \frac{1}{\cos^2 x}$$

$$\text{Let } x = a \tan t \quad (|t| \leq \frac{\pi}{2})$$

$$\sqrt{a^2 + x^2} = \sqrt{a^2(1 + \tan^2 t)} = a \sec t = \frac{a}{\cos t}$$

$$dx = a \sec^2 t dt$$

$$\int \frac{1}{\sqrt{a^2 + x^2}} dx = \int \frac{a \sec^2 t}{a \sec t} dt = \int \sec t dt$$

$$= \ln |\sec t + \tan t| + C$$

$$= \ln \left| \frac{1 + \sinh t}{1 - \sinh t} \right| + C.$$

$$\cos t = \frac{a}{\sqrt{a^2+x^2}}$$

$$\left(\sin t = \frac{x}{\sqrt{a^2+x^2}} \right) \quad 1 - \cos^2 t = \frac{x^2}{a^2+x^2}$$

$$\frac{1 + \sin t}{1 - \sin t} = \frac{1 + \frac{x}{\sqrt{a^2+x^2}}}{1 - \frac{x}{\sqrt{a^2+x^2}}} = \frac{\sqrt{a^2+x^2} + x}{\sqrt{a^2+x^2} - x}$$

$$= \frac{1}{a^2} (\sqrt{a^2+x^2} + x)^2$$

$$= \frac{1}{2} \ln \left| \frac{1}{a^2} (\sqrt{a^2+x^2} + x)^2 \right| + C.$$

$$= \ln \left(\frac{x + \sqrt{a^2+x^2}}{a} \right) + C$$

$$= \ln (x + \sqrt{a^2+x^2}) - \ln a + C$$

$$= \ln (x + \sqrt{a^2+x^2}) + C.$$

$$\textcircled{3} \int \frac{1}{\sqrt{x^2-a^2}} dx$$

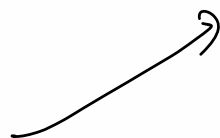
$$x = a \sec t \quad (0 < t < \frac{\pi}{2} \text{ 或 } -\frac{\pi}{2} < t < 0).$$

$$\underline{\underline{\sqrt{x^2 - a^2} = a \tan t}} \quad dx = a \sec t \tan t \, dt$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} \, dx = \int \frac{\cancel{a} \sec t \tan t}{a \tan t} \, dt$$

$$= \int \sec t \, dt = \ln | \tan t + \sec t | + C$$

$$\begin{cases} \sec t = \frac{x}{a} \\ \tan t = \frac{\sqrt{x^2 - a^2}}{a} \end{cases}$$



$$= \ln \left| \frac{\sqrt{x^2 - a^2}}{a} + \frac{x}{a} \right| + C$$

$$= \ln | \sqrt{x^2 - a^2} + x | \quad (-\ln a + C)$$

$$= \ln | \sqrt{x^2 - a^2} + x | + C.$$

例 1:

求不定积分

$$\int \frac{1}{\sqrt{a^2 - x^2}} \, dx$$

$$x = a \sin t / a \cos t$$

$$dx = a \cos t / a \cos^2 t$$

$$\left\{ \begin{array}{l} \frac{\sqrt{a^2 + x^2}}{\sqrt{x^2 - a^2}} \\ x = a \sec t / \csc t. \end{array} \right.$$

$$\sqrt{ax^2 + bx + c} \quad \int \frac{a(x-A)^2 + B}{\sqrt{\quad}} =$$

↓

$$\sqrt{a^2 - u^2}, \quad \sqrt{u^2 \pm d^2}$$

$$\cosh x = \frac{e^x + e^{-x}}{2} \quad \sinh x = \frac{e^x - e^{-x}}{2}$$

$$\cosh^2 x - \sinh^2 x = 1.$$

④ $\int \frac{1}{\sqrt{x^2 + a^2}} dx \quad x = a \sinh t \quad dx = a \cosh t dt$

$$\sqrt{x^2 + a^2} = a \cosh t.$$

$$\int a \cosh t dt \quad \int \frac{dt}{\cosh t} = t + \ln \left| \frac{e^t - 1}{e^t + 1} \right| + C$$

$$\int \frac{1}{a \cosh t} = \int \frac{1}{a e^t + a e^{-t}} = \frac{1}{2a} \ln \left(\frac{e^t - 1}{e^t + 1} \right) + C$$

$$t = \operatorname{arcsinh} \frac{x}{a} = \ln \left(\frac{x}{a} + \sqrt{\frac{x^2}{a^2} + 1} \right) + C$$

Plugging

$$= \ln \left(x + \sqrt{x^2 + a^2} \right) + C.$$

$$\textcircled{5} \int \frac{x^5}{\sqrt{1+x^2}} dx$$

$$\triangle x = \tan t \quad dx = \sec^2 t \, dt$$

$$\sqrt{1+x^2} = \sec t.$$

$$\begin{aligned} \text{解. } \int \frac{x^5}{\sqrt{1+x^2}} dx &= \int \frac{\tan^5 t}{\sec t} \sec^2 t \, dt \\ &= \int \tan^5 t \sec t \, dt \\ &= \int \tan^4 t \, \underline{\tan t \sec t} \, dt \\ &= \int \underline{\tan^4 t} \, d \sec t \end{aligned}$$

$$= \int (\sec^2 t - 1)^2 d \sec t$$

$$= \int 1 - 2 \sec^2 t + \sec^4 t d \sec t$$

$$= \frac{1}{5} \sec^5 t - \frac{2}{3} \sec^3 t + \sec t + C.$$

$$= \frac{1}{5} (1+x^2)^2 \sqrt{1+x^2} - \frac{2}{3} (1+x^2) \sqrt{1+x^2} + \sqrt{1+x^2} + C$$

$$= \frac{1}{15} (3x^4 - 4x^2 + 8) \sqrt{1+x^2} + C.$$

Ex 7.2.

$$\text{Let } t = \sqrt{1+x^2}$$

$$x^2 = t^2 - 1$$

$$2x dx = 2t dt$$

$$x dx = t dt$$

$$\int \frac{x^5}{\sqrt{1+x^2}} dx = \int \frac{x^4 \cdot x dx}{\sqrt{1+x^2}} = \int \frac{(t^2-1)^2}{t} t dt$$

$$= \int (t^2-1)^2 dt$$

$$\textcircled{6} \int \frac{1}{\sqrt{x} (1 + \sqrt[3]{x})} dx$$

$$\underline{\text{令 } t = \sqrt[6]{x}}$$

$$2\sqrt{x} \quad 3\sqrt{x} \quad 6\sqrt{x}$$

$$x = t^6$$

$$\underline{dx = 6t^5 dt}$$

$$= 6 \int \frac{1}{t^3 (1 + t^2)} t^5 dt$$

$$= 6 \int \frac{t^2}{1 + t^2} dt$$

$$= 6 \int \frac{t^2 + 1 - 1}{1 + t^2} dt$$

$$= 6 \int \left(1 - \frac{1}{1 + t^2} \right) dt$$

$$= 6 (t - \arctan t) + C$$

$$= 6 (\sqrt[6]{x} - \arctan \sqrt[6]{x}) + C.$$

$$\textcircled{7} \int \frac{\sinh \sqrt{x}}{\sqrt{x}} dx$$

$$\underline{2\sqrt{x} = (\sqrt[4]{x})'}$$

$$2 \int \sin \sqrt{x} \cdot \frac{1}{2} \frac{1}{\sqrt{x}} dx = 2 \int \sin \sqrt{x} d(\sqrt{x}),$$

$$= -2 \cos \sqrt{x} + C.$$

$$\begin{aligned} \Delta t = \sqrt{x} & \quad \begin{cases} x = t^2 \\ dx = 2t dt \end{cases} \end{aligned}$$

$$\int \frac{\sin \sqrt{x}}{\sqrt{x}} dx = \int \frac{\sin t}{t} \cdot 2t dt$$

$$= 2 \int \sin t dt$$

$$= -2 \cos t + C = -2 \cos \sqrt{x} + C.$$

$$\textcircled{8} \int \sqrt{a^2 + x^2} dx$$

$$\Delta x = a \tan t$$

$$\int a \sec t \cdot a \sec^2 t dt = \underline{\underline{a^2 \int \sec^3 t dt}}$$

$$= x \sqrt{a^2 + x^2} - \int x d(\sqrt{a^2 + x^2})$$

$$= x \sqrt{a^2 + x^2} - \int x \cdot \frac{x}{\sqrt{a^2 + x^2}} dx = \underline{\int \frac{x^2}{\sqrt{a^2 + x^2}} dx}$$

$$= x \sqrt{a^2 + x^2} - \int \frac{x^2 + a^2 - a^2}{\sqrt{a^2 + x^2}} dx$$

$$= x \sqrt{a^2 + x^2} - \underbrace{\int \sqrt{a^2 + x^2} dx} + a^2 \underbrace{\int \frac{1}{\sqrt{a^2 + x^2}} dx}$$

$$\Rightarrow \int \sqrt{a^2 + x^2} dx = \frac{x}{2} \sqrt{a^2 + x^2} + \frac{a^2}{2} \int \frac{1}{\sqrt{a^2 + x^2}} dx$$

$$= \frac{x}{2} \sqrt{a^2 + x^2} + \frac{a^2}{2} \ln (x + \sqrt{a^2 + x^2}) + C.$$

§ 有理函数的原函数

$$\underbrace{R(x) = \frac{P(x)}{Q(x)} = \frac{a_n x^n + \dots + a_0}{b_m x^m + \dots + b_0}}_{\text{rational}} = \frac{P_n(x)}{Q_m(x)}, \quad a_n, b_m \neq 0.$$

$P(x)$ 和 $Q(x)$ 没有相同的因子

$$\underbrace{P(a) = 0}_{\text{根}} \begin{cases} P(x) = (x-a) \hat{P}(x), & \downarrow n-1 \text{ 次多项式} \\ P(x) = \underbrace{(x-a)^k} \underbrace{\tilde{P}(x)}, & n-k \text{ 次多项式} \\ Q(x) = \underbrace{(x-a) \hat{Q}(x)}, \end{cases}$$

若 $n < m$, 真分式

若 $n > m$, 假分式.

$$\begin{aligned} \text{例: } \frac{x^5}{1-x^2} &= \frac{x^5 - x^3 + x^3}{1-x^2} = \frac{x^3(x^2-1) + x^3}{1-x^2} \\ &= -x^3 + \frac{x^3 - x + x}{1-x^2} \end{aligned}$$

$$= \underbrace{-x^3 - x}_{\text{多项式}} + \underbrace{\frac{x}{1-x^2}}_{\text{真分式}}$$

$$n=m.$$

a + 实数.

部分分式定理.

$$m = \alpha + \dots + \beta + 2\mu + \dots + 2\nu.$$

定理 4.3.1 设 $R(x) = \frac{P(x)}{Q(x)}$ 是一个真分式, 其分母 $Q(x)$ 具有分解式

$$Q(x) = (x-a)^\alpha \cdots (x-b)^\beta \underbrace{(x^2+px+q)^\mu \cdots (x^2+rx+s)^\nu}_{\alpha + \dots + \beta + 2\mu + \dots + 2\nu},$$

其中 $a, \dots, b, p, q, \dots, r, s$ 为实数, 且 $p^2 - 4q < 0, \dots, r^2 - 4s < 0$, $\alpha, \dots, \beta, \mu, \dots, \nu$ 为自然数, 则

$$\begin{aligned} R(x) = & \underbrace{\frac{A_\alpha}{(x-a)^\alpha} + \frac{A_{\alpha-1}}{(x-a)^{\alpha-1}} + \cdots + \frac{A_1}{x-a}}_{\alpha \text{ 项}} + \underbrace{\frac{B_\beta}{(x-b)^\beta} + \frac{B_{\beta-1}}{(x-b)^{\beta-1}} + \cdots + \frac{B_1}{x-b}}_{\beta \text{ 项}} \\ & + \underbrace{\frac{K_\mu x + L_\mu}{(x^2+px+q)^\mu} + \cdots + \frac{K_1 x + L_1}{x^2+px+q}}_{\mu \text{ 项}} + \cdots \\ & + \frac{M_\nu x + N_\nu}{(x^2+rx+s)^\nu} + \cdots + \frac{M_1 x + N_1}{x^2+rx+s}, \end{aligned}$$

其中 $A_i, \dots, B_i, K_i, L_i, \dots, M_i, N_i$ 都是实常数, 并且这个分解式的系数都是惟一确定的.

$$\left(\frac{A}{(x-a)^k}, \frac{Ax+B}{(x^2+px+q)^k} \right)$$

$$p^2 - 4q < 0, \quad k=1, 2, 3, \dots$$

$$\textcircled{1} \text{ 合解: } \frac{x+1}{x^2-4x+3}$$

$\mathbb{Q}(x)$

$$x^2-4x+3 = \underline{(x-1)(x-3)}$$

$$\frac{x+1}{x^2-4x+3} = \frac{A}{x-1} + \frac{B}{x-3} = -\frac{1}{x-1} + \frac{2}{x-3}$$

$$x+1 = A(x-3) + B(x-1)$$

① 代入法: 代入 $x=1$

$$2 = -2A \Rightarrow A = -1$$

代入 $x=3$

$$4 = 2B \Rightarrow B = 2$$

$$\begin{cases} A+B=1 \\ 3A+B=-3 \end{cases}$$

$$\textcircled{2} \text{ 合解: } \frac{x}{x^3+x^2+3x+3} = \mathbb{Q}(x)$$

$$\begin{aligned} & x^2(x+1) + 3(x+1) \\ &= \underline{(x+1)} \underline{(x^2+3)} \end{aligned} \quad \begin{matrix} -12 \\ p^2-4q < 0 \end{matrix}$$

$$\frac{x}{x^3+x^2+3x+3} = \frac{A}{x+1} + \frac{Bx+C}{x^2+3}$$

$$x = \frac{A(x^2+3)}{x^2+3} + \frac{(Bx+C)(x+1)}{x^2+3}$$

代入 $x=-1 \Rightarrow -1 = 4A \Rightarrow A = -\frac{1}{4}$

$$\frac{1}{4}x^2 + x + \frac{3}{4} = Bx^2 + (B+C)x + C$$

$$B = \frac{1}{4} \quad C = \frac{3}{4}$$

③ 分解: $\frac{x^3+1}{x^4-3x^3+3x^2-x} = Q(x)$

$$Q(x) = x(x^3-3x^2+3x-1)$$

$$= \boxed{x} (x-1)^3$$

$$\frac{x^3+1}{x^4-3x^3+3x^2-x} = \frac{-1}{x} + \frac{2}{(x-1)^3} + \frac{1}{(x-1)^2} + \frac{2}{x-1}$$

$$x^3+1 = A(x-1)^3 + Bx + Cx(x-1) + Dx(x-1)^2$$

代入 $x=0$ $1 = -A \Rightarrow A = -1$

$$2x^2 - 3x + 1 = B + C(x-1) + D(x-1)^2$$

let $x=1$

$$2 = B$$

$$2x^2 - 3x + 1 = \underline{C(x-1) + D(x-1)^2}$$

$$C=1 \quad D=2$$

$$\underline{4x-3 = 2D(x-1) + C.}$$

$$\underline{\int \frac{A}{(x-a)^k} dx}, \quad \boxed{\int \frac{Ax+B}{(x^2+px+q)^k} dx} \quad \begin{cases} p^2-4q < 0 \\ k \geq 1 \end{cases}$$

$$\textcircled{1} \int \frac{A}{x-a} dx = A \int \frac{1}{x-a} dx = A \ln |x-a| + C.$$

$$\textcircled{2} \int \frac{1}{(x-a)^k} dx = \int \frac{1}{(x-a)^k} d(x-a) = \frac{1}{1-k} (x-a)^{1-k} + C.$$

$$q - \frac{p^2}{4} > 0.$$

$$\underline{x^2 + px + q = \left(x + \frac{p}{2}\right)^2 + \left(q - \frac{p^2}{4}\right)}$$

$$\Delta a^2 = q - \frac{p^2}{4}$$

$$= \left(x + \frac{p}{2}\right)^2 + a^2 \quad \underline{\Delta u = x + \frac{p}{2}} \quad du = dx$$

$$\underline{\int \frac{Ax+B}{(x^2+px+q)^k} dx = \int \frac{Au + (B - \frac{Ap}{2})}{(u^2 + a^2)^k} du}$$

$$= \underbrace{A \int \frac{u}{(u^2+a^2)^k} du} + \underbrace{(B - \frac{Ap}{2}) \int \frac{1}{(u^2+a^2)^k} du}$$

$d(u^2+a^2) = 2 du$

$$\textcircled{3} \quad k=1 \text{ of.} \quad \underline{\int \frac{u}{u^2+a^2} du = \frac{1}{2} \int \frac{2u}{u^2+a^2} du = \frac{1}{2} \int \frac{1}{u^2+a^2} d(u^2+a^2)}$$

$$= \underline{\frac{1}{2} \ln(u^2+a^2) + C.}$$

$$\underline{\int \frac{1}{u^2+a^2} du = \frac{1}{a} \arctan \frac{u}{a} + C.}$$

$$\int \frac{Ax+B}{x^2+px+q} dx = \frac{1}{2} A \ln \left(\left(q - \frac{p^2}{4}\right) + \left(x + \frac{p}{2}\right)^2 \right)$$

$$+ \frac{B - \frac{Ap}{2}}{\sqrt{q - \frac{p^2}{4}}} \arctan \left(\frac{x + \frac{p}{2}}{\sqrt{q - \frac{p^2}{4}}} \right) + C$$

④ $k \geq 2$ 时.

$$\int \frac{u}{(u^2+a^2)^k} du$$

$$\int \frac{1}{(u^2+a^2)^k} du$$

$$\frac{1}{2} \int \frac{1}{(u^2+a^2)^k} d(u^2+a^2) = \frac{1}{2(1-k)} (u^2+a^2)^{1-k} + C$$

$$\text{记 } I_k = \int \frac{1}{(u^2+a^2)^k} du$$

$$-k \cdot (u^2+a^2)^{-k-1} \cdot 2u$$

$$I_k = \frac{u}{(u^2+a^2)^k} - \int u d \left(\frac{1}{(u^2+a^2)^k} \right)$$

$$- \frac{k}{(u^2+a^2)^{k+1}} \cdot 2u$$

$$= \frac{u}{(u^2+a^2)^k} + 2k \int \frac{u^2}{(u^2+a^2)^{k+1}} du$$

$$= \frac{u}{(u^2+a^2)^k} + 2k \int \frac{u^2+a^2-a^2}{(u^2+a^2)^{k+1}} du$$

$$= \frac{u}{(u^2+a^2)^k} + 2k \underbrace{\int \frac{1}{(u^2+a^2)^k} du}_{I_k} - 2k a^2 \underbrace{\int \frac{1}{(u^2+a^2)^{k+1}} du}_{I_{k+1}}$$

$$I_k = \frac{u}{(u^2+a^2)^k} + 2k I_k - 2k a^2 \underline{\underline{I_{k+1}}}$$

$$\boxed{I_{k+1} = \frac{1}{2ka^2} \frac{u}{(u^2+a^2)^k} + \frac{2k-1}{2ka^2} I_k.}$$

$$\left\{ \begin{array}{l} I_1 = \frac{1}{a} \arctan \frac{u}{a} + C. \end{array} \right.$$

$$I_2 = \frac{1}{2a^2} \frac{u}{a^2+u^2} + \frac{1}{2a^2} I_1$$

$$I_3 = \frac{1}{4a^2} \frac{u}{(u^2+a^2)^2} + \frac{3}{4a^2} I_2$$

$$\left\{ \begin{array}{l} u = \frac{p}{2} + x \\ a^2 = \sqrt{q - \frac{p^2}{4}} \end{array} \right.$$

$$\textcircled{4} \int \frac{x^3+1}{x^4-3x^3+3x^2-x} dx$$

$$= - \int \frac{1}{x} dx + 2 \int \frac{dx}{(x-1)^3} + \int \frac{1}{(x-1)^2} dx + 2 \int \frac{1}{x-1} dx$$

$$= \underbrace{-\ln|x|} - \underbrace{\frac{1}{(x-1)^2} - \frac{1}{x-1}} + 2 \ln|x-1| + C$$

$$= \ln \frac{(x-1)^2}{|x|} - \frac{x}{(x-1)^2} + C.$$

$$\textcircled{5} \int \frac{5x+6}{\underbrace{x^2+x+1}} dx \quad p^2-4q = -3 < 0.$$

$$(x^2+x+1)' = \underline{\underline{2x+1}}$$

$$5x+6 = \frac{5}{2} (2x+1) + \frac{7}{2}$$

$$\int \frac{5x+6}{x^2+x+1} dx = \frac{5}{2} \int \frac{(x^2+x+1)'}{x^2+x+1} dx + \frac{7}{2} \int \frac{1}{x^2+x+1} dx$$

$$= \underline{\underline{\frac{5}{2} \ln(x^2+x+1) + \frac{7}{2} \int \frac{1}{x^2+x+1} dx}}$$

$$\int \frac{1}{x^2+x+1} dx = \int \frac{dx}{(x+\frac{1}{2})^2 + (\frac{\sqrt{3}}{2})^2} = \int \frac{dy}{u^2 + (\frac{\sqrt{3}}{2})^2}$$

$$u = x + \frac{1}{2} \quad dx = du$$

$$= \frac{7}{\sqrt{3}} \arctan \frac{2x+1}{\sqrt{3}} + C.$$

$$\textcircled{6} \int \frac{5x+3}{(x^2-2x+5)^2} dx$$

$$x^2 - 2x + 5 = (x-1)^2 + 2^2$$

$$(x^2 - 2x + 5)' = 2x - 2$$

$$5x + 3 = \frac{5}{2}(2x - 2) + 8.$$

$$\int \frac{5x+3}{(x^2-2x+5)^2} dx = \frac{5}{2} \int \frac{(x^2-2x+5)'}{(x^2-2x+5)^2} dx + \int \frac{8}{(x^2-2x+5)^2} dx$$

$$= -\frac{5}{2} \frac{1}{x^2-2x+5} + 8 \int \frac{1}{(x-1)^2+2^2} dx$$

$$u = x-1$$

$$a = 2.$$

$$I_2 = \frac{1}{2a^2} \left(\frac{u}{a^2 + u^2} + I_1 \right) = \frac{1}{8} \left(\frac{(x-1)^2}{2^2 + (x-1)^2} + I_1 \right)$$

$$I_1 = \frac{1}{a} \arctan \frac{u}{a} + C = \frac{1}{2} \arctan \frac{x-1}{2} + C$$

$$= -\frac{5}{2} \frac{1}{x^2 - 2x + 5} + \frac{x-1}{x^2 - 2x + 5} + \frac{1}{2} \arctan \left(\frac{x-1}{2} \right) + C$$

$$= \frac{2x-7}{2(x^2 - 2x + 5)} + \dots$$

⑦. $\int \frac{1}{x^4 (1+x^2)} dx \quad \text{令 } x = \frac{1}{t}$

$$dx = -\frac{1}{t^2} dt$$

$$= \int \frac{1}{\left(\frac{1}{t}\right)^4 (1 + \left(\frac{1}{t}\right)^2)} \cdot -\frac{1}{t^2} dt$$

$$= \int \frac{t^4 \cdot \cancel{t^2}}{1+t^2} \cdot -\frac{1}{\cancel{t^2}} dt$$

$$= - \int \frac{(t^4 - 1)}{1+t^2} dt = - \int (t^2 - 1) dt - \int \frac{1}{1+t^2} dt$$

$$= -\frac{1}{3}t^3 + t - \arctan t + C$$

$$= \frac{1}{x} - \frac{1}{3} \frac{1}{x^3} - \arctan \frac{1}{x} + C.$$

三角代换

$$\begin{matrix} \sin x & \cos x \\ R(\sin x, \cos x) \end{matrix}$$

$$\begin{aligned} & \int \sin^5 x \cos^2 x dx \\ & \int \frac{1}{\cos x} dx \quad \int \sec^n x dx \end{aligned}$$

$$\left\{ \begin{aligned} \sin \alpha &= \frac{2 \tan \frac{\alpha}{2}}{1 + \tan^2 \frac{\alpha}{2}} \end{aligned} \right.$$

$$\cos \alpha = \frac{1 - \tan^2 \frac{\alpha}{2}}{1 + \tan^2 \frac{\alpha}{2}}$$

$$\tan \alpha =$$

$$\hat{=} \underline{t = \tan \frac{x}{2}} \quad (|x| < \pi)$$

$$\sinh x = \frac{2t}{1+t^2}$$

$$\cosh x = \frac{1+t^2}{1-t^2}$$

$$x = 2 \operatorname{arctan} t$$

$$dx = \frac{2}{1+t^2} dt$$

$$\int R(\sinh x, \cosh x) dx = \int R\left(\frac{2t}{1+t^2}, \frac{1-t^2}{1+t^2}\right) \cdot \frac{2}{1+t^2} dt$$

$$\textcircled{1} \int \frac{1}{\sinh x \cosh x} dx = \int \frac{1}{\left(\frac{2t}{1+t^2}\right) \left(1 + \frac{1-t^2}{1+t^2}\right)} \cdot \frac{2}{1+t^2} dt$$

$$\textcircled{2} \hat{=} t = \tan \frac{x}{2}$$

$$= \int \frac{1+t^2}{2t} \cdot \frac{1+t^2}{2} \cdot \frac{2}{1+t^2} dt$$

$$= \frac{1}{2} \int \left(\frac{1}{t} + t \right) dt$$

$$= \frac{1}{4} t^2 + \frac{1}{2} \ln|t| dt$$

$$= \frac{1}{4} \tan^2 \frac{x}{2} + \frac{1}{2} \ln \left| \tan \frac{x}{2} \right| + C.$$

Ex. $\sqrt{1-t} = \cos x$

$$\int \frac{1}{\cos x} dx \quad \int \frac{\cos x}{\cos^2 x} dx \quad \int \frac{d \sin x}{1 - \sin^2 x}$$

$$\int \frac{1}{\sin x (1 + \cos x)} dx$$

$$= \int \frac{\sin x}{\sin^2 x (1 + \cos x)} dx$$

$$= - \int \frac{d \cos x}{(1 - \cos^2 x) (1 + \cos x)}$$

$$= - \int \frac{1}{\dots} dt$$

$$= -\frac{1}{2} \int \frac{1}{1+t} \left(\frac{1}{1-t} + \frac{1}{1+t} \right) dt$$

$$= -\frac{1}{2} \int \frac{1}{(1+t)^2} dt - \frac{1}{2} \int \frac{1}{1+t} \cdot \frac{1}{1-t} dt$$

$$= \frac{1}{2} \frac{1}{1+t} - \frac{1}{4} \int \left(\frac{1}{1+t} + \frac{1}{1-t} \right) dt$$

$$= \frac{1}{2} \frac{1}{1+t} + \frac{1}{4} \ln \left| \frac{1-t}{1+t} \right| + C.$$

$$= \frac{1}{2(1+\cos x)} + \frac{1}{4} \ln \left| \frac{1-\cos x}{1+\cos x} \right| + C$$

$$\textcircled{5} \int \frac{1}{\sin^4 x} dx$$

$$\textcircled{A} \quad \text{Let } t = \tan \frac{x}{2}$$

$$\sin x = \frac{2t}{1+t^2}$$

$$\int \frac{1}{\sinh^4 x} dx = \int \left(\frac{1+t^2}{2t} \right)^4 \cdot \frac{2}{1+t^2} dt$$

$$= \int \frac{t^6 + 3t^4 + 3t^2 + 1}{8t^4} dt$$

$$= \frac{1}{8} \left(\frac{1}{3}t^3 + 3t - \frac{3}{t} - \frac{1}{3t^3} \right) + C.$$

$$\sec^2 x = 1 + \tan^2 x$$

$$\textcircled{=} \quad \wedge \quad t = \tan x$$

$$\cos x = \frac{1}{\sec x} = \frac{1}{\sqrt{1+\tan^2 x}} = \frac{1}{\sqrt{1+t^2}}$$

$$\sinh x = \tan x \cdot \cos x = \frac{t}{\sqrt{1+t^2}}$$

$$dx = \text{derivative} = \frac{1}{1+t^2}$$

$$\int \frac{1}{\sinh^4 x} dx = \int \left(\frac{\sqrt{1+t^2}}{t} \right)^4 \cdot \frac{1}{1+t^2} dt$$

$$= \int \frac{(1+t^2)^{\frac{3}{2}}}{t^4} \cdot \frac{1}{1+t^2} dt = \int \frac{1+t^2}{t^4} dt$$

$$= -\frac{1}{3t^3} - \frac{1}{t} + C$$

$$= -\frac{1}{3\tan^3 x} - \frac{1}{\tan x} + C.$$

③. $\frac{1}{\sin x} = \boxed{\csc x}$

$$\underline{\underline{\int \csc^4 x dx}}$$

$$\frac{1}{\sin^2 x} = \csc^2 x = 1 + \cot^2 x$$

$$\begin{aligned} (\cot x)' &= -\csc^2 x \\ &= -\frac{1}{\sin^2 x} \end{aligned}$$

$$\int \frac{1}{\sin^2 x} \cdot \frac{1}{\sin^2 x} dx = \int \csc^2 x (1 + \cot^2 x) dx$$

$$= \int \csc^2 x dx + \int \csc^2 x \cot^2 x dx$$

$$= \underline{\underline{\int \csc^2 x \, dx}} - \underline{\underline{\int \cot^2 x \, d \cot x}}$$

$$= -\cot x - \frac{1}{3} \cot^3 x + C.$$

$$\boxed{n \sqrt{ax+b}}$$

$$t = \boxed{n \sqrt{\frac{ax+b}{cx+d}}}$$

$$\boxed{\int \frac{x^5}{\sqrt{1+x^2}} \, dx}$$

$$t = \sqrt{1+t^2}$$

$$\underline{\underline{\int \frac{1}{\sqrt{x}(1+\sqrt{x})} \, dx}}$$

$$\text{let } t = \sqrt{x}$$

$$\textcircled{7} \int \frac{1}{\sqrt[4]{(x-2)^3 (x+1)^5}} \, dx$$

$$= \int \frac{1}{\sqrt[4]{\underbrace{(x-2)^4}_{(x-2)} \cdot \underbrace{(x+1)^4}_{(x+1)} \cdot (x+1)}} \, dx = \int \frac{1}{(x-2)(x+1)} \cdot \sqrt[4]{\frac{x-2}{x+1}} \, dx$$

$$\triangle t = \sqrt[4]{\frac{x-2}{x+1}} \quad t^4 = \frac{x-2}{x+1}$$

$$dx = \frac{12t^3}{(1-t^4)^2} dt$$

$\Leftarrow$

$$x = \frac{t^4 + 2}{1 - t^4}$$

$$= \int \frac{1}{\left(\frac{t^4+2}{1-t^4} - 2\right) \left(\frac{t^4+2}{1-t^4} + 1\right)} \cdot t \cdot \frac{12t^3}{(1-t^4)^2} dt$$

$$= \int \frac{4}{3} dt = \frac{4}{3} t + C.$$

$$= \frac{4}{3} \sqrt[4]{\frac{x-2}{x+1}} + C.$$

$$\int \frac{Ax+B}{ax^2+bx+c} dx$$

$$\int \sqrt{a^2 - u^2} du$$

$$\int \frac{Ax+B}{\sqrt{ax^2+bx+c}} dx$$

$$\int (Ax+B) \sqrt{ax^2+bx+c} dx$$

$$(ax^2+bx+c)' = \underline{\underline{2ax+b}}$$

$$(Ax+B) = \frac{A}{2a} (2ax+b) + (B - \frac{Ab}{2a}) \quad \int \frac{1}{\sqrt{x}} dx \quad 2\sqrt{x}$$

$$= \frac{A}{2a} \int \frac{d(ax^2+bx+c)}{\sqrt{ax^2+bx+c}} + \left(B - \frac{Ab}{2a}\right) \int \frac{1}{\sqrt{ax^2+bx+c}} dx$$

$$= \frac{A}{a} \sqrt{ax^2+bx+c} +$$

$$\underline{\underline{b^2 - 4ac < 0}}$$

$$\int \frac{1}{\sqrt{a\left(x + \frac{b}{2a}\right)^2 + \left(c - \frac{b^2}{4a}\right)}} dx$$

$$u = x + \frac{b}{2a}$$

$$\int \frac{1}{\sqrt{u^2 \pm p^2}} du$$

$$\int \frac{1}{\sqrt{u^2 \pm p^2}} du$$

$$\int \sqrt{p^2 - u^2}$$

§ 定积分.

① 曲边梯形的面积.

$f(x)$ 在 $[a, b]$ 上 $f(x) > 0$.

$$y = ax + b$$

分割:

$$a = x_0 < x_1 < x_2 < \dots < x_n = b$$

$$I_i = [x_{i-1}, x_i]$$

$$i = 1, 2, \dots, n$$

$$\Delta_i = x_i - x_{i-1}$$

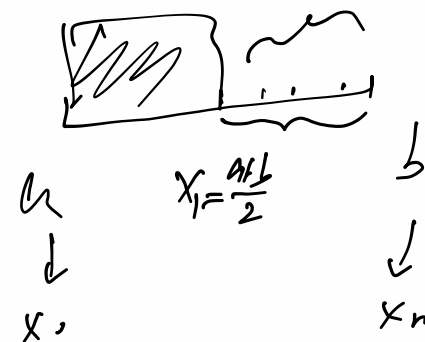
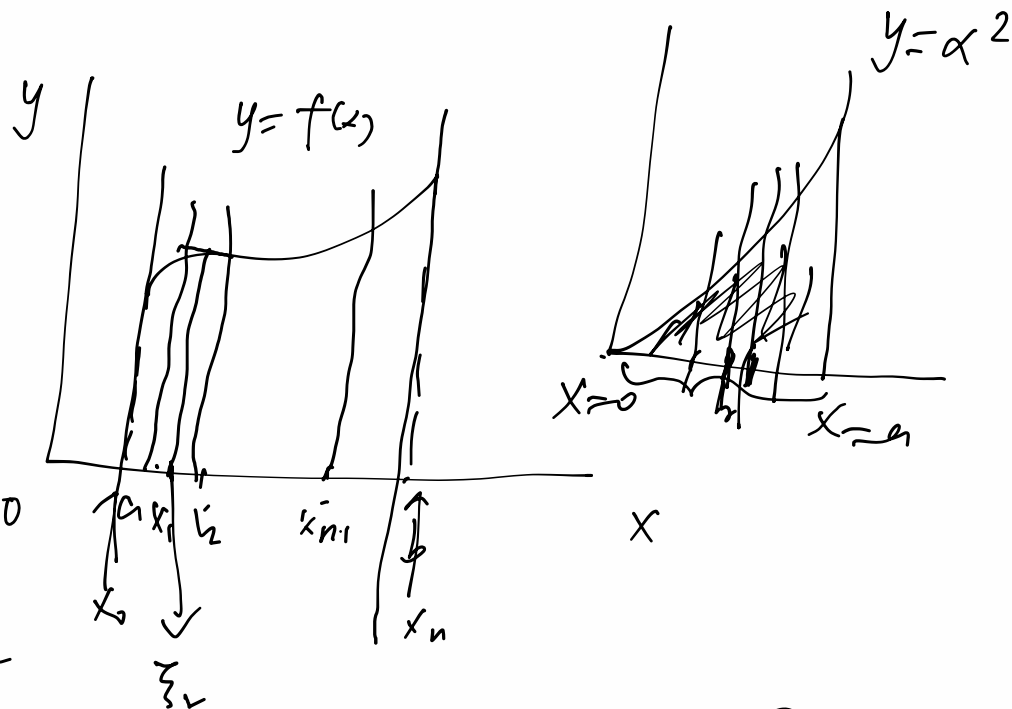
$$f(\xi_i) \Delta x_i \sim S_i$$

$$S = \sum_{i=1}^n S_i \sim \sum_{i=1}^n f(\xi_i) \Delta x_i$$

$$\lambda \rightarrow 0 \quad n \rightarrow \infty$$

令 $\lambda := \max \Delta x_i$ 宽度 $\lambda \rightarrow 0$.

$$\lim_{\lambda \rightarrow 0} \sum_{i=1}^n f(\xi_i) \Delta x_i = \textcircled{I}$$



② 变元在运动,

$v(t)$ 为 $t=a$ 到 $t=b$ 的路程

$$\frac{f(t)}{f'(t)=v(t)} \quad \frac{S(t)=f(t) \text{ (圈)}}{S(b)-S(a)}$$

$$\underline{v(t) \equiv v.}$$

$$S = v(b-a).$$

变元运动. $[a, b].$

$$t_0 < t_1 < t_2 < \dots < t_n$$

$\downarrow$ $\downarrow$ $\downarrow$
 a b

$$\underline{t_i \in [t_{i-1}, t_i]} \sim \xi_i \quad v(\xi_i)$$

$$\Delta S_i \sim v(\xi_i) \Delta t_i$$

$$S = \sum_{i=1}^n \Delta S_i \sim \sum_{i=1}^n v(\xi_i) \Delta t_i$$

$$\lambda = \max \Delta t_i \quad \lambda \rightarrow 0.$$

定积分的定义:

设 $f(x)$ 在 $[a, b]$ 上有定义: 分割 P 为:

$$a = x_0 < x_1 < x_2 < \dots < x_{n-1} < x_n = b.$$

$$\Delta x_i = x_i - x_{i-1} \quad (i=1, \dots, n). \quad \lambda := \max \Delta x_i$$

任取 $\xi_i \in [x_{i-1}, x_i]$. $\{x_i\}_0^n$ 分点序列

$\{\xi_i\}_1^n$ 任点序列

$$\lim_{\lambda \rightarrow 0} \sum_{i=1}^n f(\xi_i) \Delta x_i = \int$$

记 I 为 $f(x)$ 在

$[a, b]$ 上的定积分.

黎曼和.

(5.5) 设 $f(x)$ 在 $[a, b]$ 上连续, I 是一个实数. 如果 $\forall \varepsilon > 0, \exists \delta > 0$.

s.t. 任给一个分点 $\{x_i\}$. 和任给任点 $\{\xi_i\}$ 只要满足

$\lambda < \delta$. 那么都有

$$\left| \sum_{i=1}^n f(\xi_i) \Delta x_i - I \right| < \varepsilon.$$

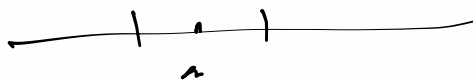
(b) 黎曼和

$$f(x) dx = I = \lim_{\lambda \rightarrow 0} \sum_{i=1}^n f(\xi_i) \Delta x_i$$

① 4x3TP6

$$\int_a^b f(x) dx = \int_a^b f(t) dt = \int_a^b f(u) du$$

① 在 $[a, b]$ 上可积.



例: Dirichlet 函数

$$D(x) = \begin{cases} 1 & x \text{ 是有理数} \\ 0 & x \text{ 是无理数} \end{cases}$$

在 $[0, 1]$ 上.

$x_0, \dots, x_n$

$[x_{i-1}, x_i]$ $\Delta x_i = x_i - x_{i-1}$

$\xi_i \in [x_{i-1}, x_i]$ $D(\xi_i) = 1$

$$\sum D(\xi_i) \Delta x_i = \sum 1 \cdot \Delta x_i = 1$$

ξ_i

$$D(\xi_i) = 0$$

结论:

$$\sum D(\xi_i) \Delta x_i = \sum 0 \cdot \Delta x_i = 0$$

结论: 若 $f(x)$ 在 $[a, b]$ 上可积, 那么 $f(x)$ 在 $[a, b]$ 上连续.

$$\exists M, \text{ s.t. } |f(x)| \leq M.$$

2.10: $\int_a^b f(x) dx = I = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(\xi_i) \Delta x_i$

$$\forall \varepsilon > 0 \exists \delta.$$

$$\left| \sum_{i=1}^n f(\xi_i) \Delta x_i - I \right| < \varepsilon$$

~~For~~ $\varepsilon = 1$.

$$\left| \sum_{i=1}^n f(\xi_i) \Delta x_i \right| < |I| + 1$$

$$\xi_i \in [x_{i-1}, x_i]$$

$$\left(f(\xi_1) \Delta x_1 + \sum_{i=2}^n f(\xi_i) \Delta x_i \right)$$

$$\left| f(\xi_1) \Delta x_1 \right| < |I| + 1 + \left| \sum_{i=2}^n f(\xi_i) \Delta x_i \right|$$

$$\xi_2, \dots, \xi_n \in \frac{x_2 + x_1}{2}, \dots, \xi_n = \frac{x_n + x_{n-1}}{2}$$

$$\xi_i \in [x_{i-1}, x_i] \quad \left| \int_{x_{i-1}}^{x_i} f(x) dx \right| < C.$$

$$\int_a^b \frac{1}{x} dx \quad (x_1, x_2) \in [a, b] \subseteq \mathbb{R}^+ \text{ s.t. } x_1 < x_2$$

定理 (5.1.2)

若 $f(x)$ 在 $[a, b]$ 上有界, 且 $\rightarrow$ 分段连续

① 只有有限个间断点

或 ② 单调函数

$\Rightarrow f(x)$ 可积, $\int_a^b f(x) dx = I$ 存在.

例: 计算 $\int_a^b x dx$

解:

$\{x_i\}$

$[x_{i-1}, x_i]$
 ξ_i

$\sum_{i=1}^n f(\xi_i) \Delta x_i$

取 $\eta_i = \frac{x_{i-1} + x_i}{2}, i=1, 2, \dots, n$

$f(x) = x$

$\sum_{i=1}^n \xi_i \Delta x_i$

$\sum_{i=1}^n \eta_i \Delta x_i$

$\sum_{i=1}^n (\xi_i - \eta_i) \Delta x_i$

$$\sum_{i=1}^n \frac{x_i + x_{i-1}}{2} (x_i - x_{i-1}) = \sum_{i=1}^n \left(\frac{1}{2} \right) (x_i^2 - x_{i-1}^2)$$

$$= \frac{1}{2} (x_n^2 - x_0^2) = \frac{1}{2} (b^2 - a^2)$$

$$\xi_i, \eta_i \in [x_{i-1}, x_i]$$

$$|\xi_i - \eta_i| \leq \Delta x_i \leq \lambda = \max \Delta x_i$$

$$\left| \sum_{i=1}^n (\xi_i - \eta_i) \Delta x_i \right| \leq \sum_{i=1}^n |\xi_i - \eta_i| \Delta x_i$$

$$\leq \lambda \sum_{i=1}^n \Delta x_i = \lambda (b-a)$$

$$\left| \sum_{i=1}^n f(\xi_i) \Delta x_i - \frac{b^2 - a^2}{2} \right| \leq \lambda (b-a)$$

$$\forall \varepsilon > 0, \exists \delta = \frac{\varepsilon}{b-a}, \quad \lambda < \frac{\varepsilon}{b-a} \leq \varepsilon.$$

$$\lim_{\lambda \rightarrow 0} \sum f(\xi_i) \Delta x_i = \frac{b^2 - a^2}{2} \quad \int_a^b x dx = \frac{b^2 - a^2}{2}$$

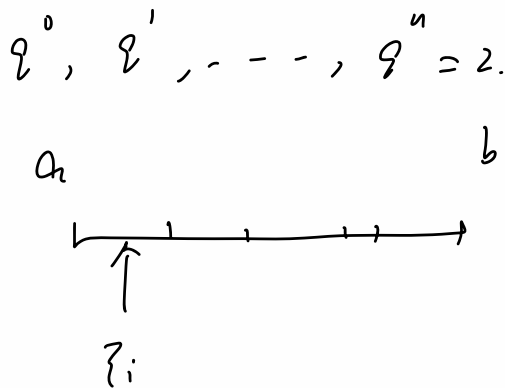
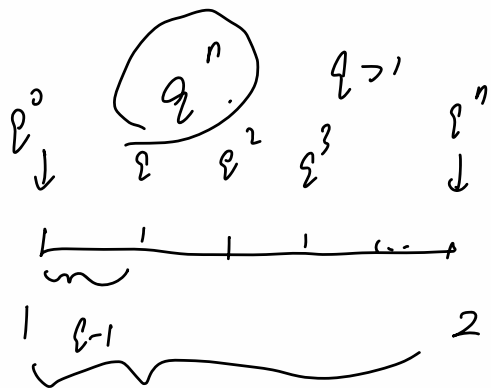
$$\int_a^b x^2 dx$$

$$\eta_i = \sqrt{\frac{x_1^2 + x_i x_{i-1} + x_{i-1}^2}{3}}$$

例: $f(x)$ 在 $[a, b]$ 上连续, $\Rightarrow f(x)$ 可积.

$$\lim_{\lambda \rightarrow 0} f(\xi_i) \Delta x_i = I$$

可以取特殊的分割点 ξ_i



$$\Delta x_i = \frac{b-a}{n}$$

$$\xi_i = a + \frac{i}{n} (b-a), \quad i=1, 2, \dots, n$$

$$\int_a^b f(x) dx = \lim_{n \rightarrow \infty} \sum_{i=1}^n f\left(a + \frac{i}{n} (b-a)\right) \frac{b-a}{n}$$

例: $\int_0^1 e^x dx$

$$\Delta x_i = \frac{1}{n}$$

$$\xi_i = \frac{i}{n}, \quad i=1, \dots, n$$

$$= \lim_{n \rightarrow \infty} \sum_{i=1}^n e^{\frac{i}{n}} \left(\frac{1}{n}\right)$$

$$= \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i=1}^n e^{\frac{i}{n}}$$

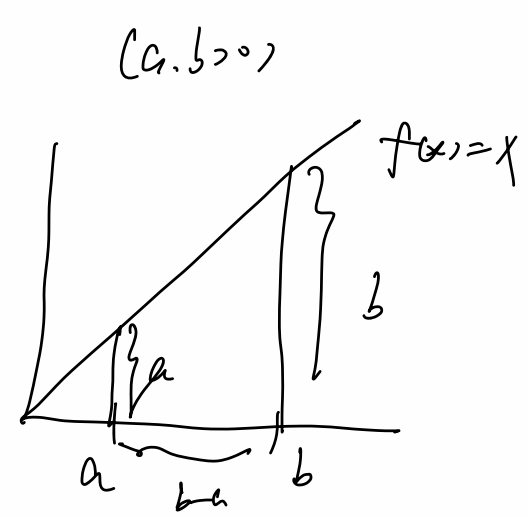
$$= \lim_{n \rightarrow \infty} \frac{1}{n} \frac{1 - (e^{\frac{1}{n}})^n}{1 - e^{\frac{1}{n}}}$$

$$e^{\frac{1}{n}} + (e^{\frac{1}{n}})^2 + (e^{\frac{1}{n}})^3 + \dots + (e^{\frac{1}{n}})^n$$

$$\frac{1 - e}{1 - e^{\frac{1}{n}}}$$

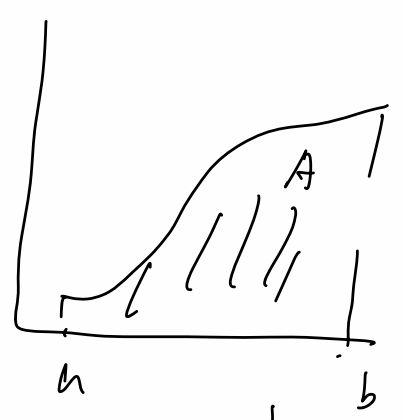
$$\lim_{n \rightarrow \infty} \frac{1}{n} \frac{1 - e}{1 - e^{\frac{1}{n}}}$$

$$\int_a^b x dx = \frac{b^2 - a^2}{2}$$



$f(x) \geq 0$

$$\int_a^b f(x) dx = A$$

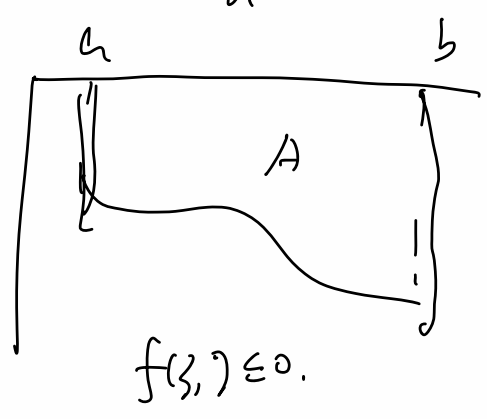


$\nexists f(x) \leq 0$

$A > 0$

$$\int_a^b f(x) dx \geq 0$$

$$\int_a^b f(x) dx = -A$$



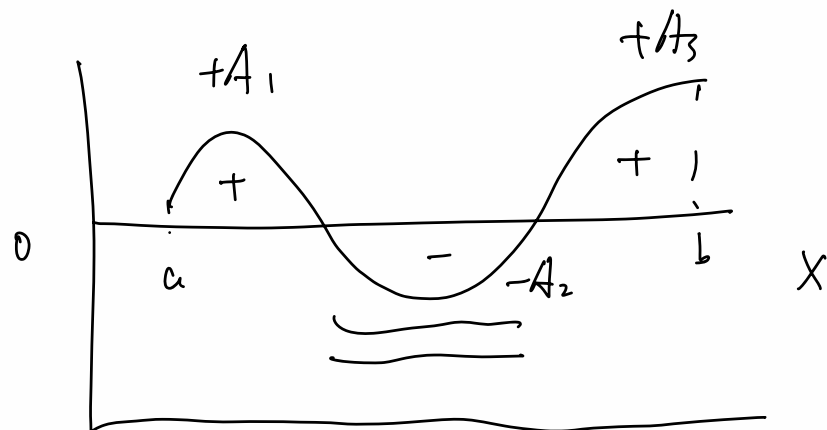
$$\int_a^b f(x) dx = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(\xi_i) \Delta x_i = \lim_{n \rightarrow \infty} - \sum_{i=1}^n |f(\xi_i)| \Delta x_i$$

$$\int_a^b f(x) dx \quad \lambda \rightarrow \infty \quad \text{if } \lambda \neq 1$$

$$\lambda \rightarrow \infty \quad \text{if } \lambda = 1$$

$$= - \int_a^b [f(x)] dx$$

$$= - \int_a^b |f(x)| dx$$



$$\int_a^b f(x) dx = A_1 - A_2 + A_3$$

$$\int_a^b |f(x)| dx$$

$$\int_a^b f(x) dx$$

$$a < b.$$

$$\int_a^a f(x) dx = 0.$$

$$\int_b^a f(x) dx = - \int_a^b f(x) dx$$

1.2 定理 (Newton-Leibniz 公式)

若 $f(x)$ 在 $[a, b]$ 上可积, 存在 $F(x)$ ① 在 $[a, b]$ 上连续

② 在 (a, b) 内可导

③ $F'(x) = f(x) \quad \forall x \in (a, b)$

则 $\int_a^b f(x) dx = \underline{\underline{F(b) - F(a)}}$ ① ($F(x)$ 与 $f(x)$ 在 (a, b) 上可积)

证明: 将 $[a, b]$ 分成 n 等份, $x_0 < x_1 < \dots < x_n$

$$\Delta x_i = \frac{b-a}{n}$$

$$\boxed{F(b) - F(a)} = \sum_{i=1}^n [F(x_i) - F(x_{i-1})]$$



Lagrange 定理

$$= \sum_{i=1}^n \underline{\underline{F'(\xi_i) \Delta x_i}} \quad \text{① } \xi_i \in (x_{i-1}, x_i)$$

$$= \underline{\underline{\sum_{i=1}^n f(\xi_i) \Delta x_i}} \quad \underline{\underline{\xi_i \in (a, b)}}$$

当 $n \rightarrow \infty, \Delta x_i \rightarrow 0$

$$F(b) - F(a) = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(\xi_i) \Delta x_i = \int_a^b f(x) dx$$

$$\underline{F(x) \Big|_a^b} = F(b) - F(a)$$

$$\underline{\int_a^b f(x) dx} = F(x) \Big|_a^b$$

$$\int_a^b dF(x) = \underline{\int_a^b f(x) dx} \Big|_a^b$$

$$\underline{\int_a^b x dx} = \underline{\frac{1}{2} x^2} \Big|_a^b = \frac{1}{2} (b^2 - a^2)$$

$$\int_a^b \sin x dx = (-\cos x) \Big|_a^b = \cos a - \cos b$$

$$\int_a^b \cos x dx = (\sin x) \Big|_a^b = \sin b - \sin a$$

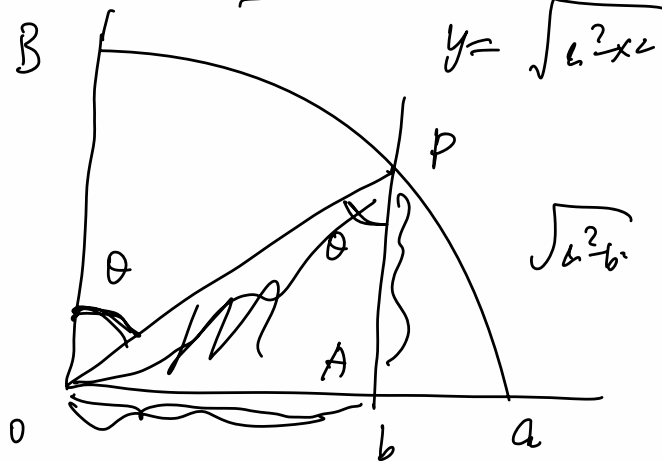
13.1 5.1.2. $\int_0^b \sqrt{a^2 - x^2} dx \quad (\underline{a \geq b > 0})$

$$\int \sqrt{a^2 - x^2} dx = \frac{a^2}{2} \arcsin \frac{x}{a} + \frac{1}{2} x \sqrt{a^2 - x^2} + C.$$

$$= \left(\frac{a^2}{2} \arcsin \frac{x}{a} + \frac{1}{2} x \sqrt{a^2 - x^2} \right) \Big|_0^b$$

$$= \frac{a^2}{2} \arcsin \frac{b}{a} + \frac{1}{2} b \sqrt{a^2 - b^2}$$

$$y = \sqrt{a^2 - x^2}$$



$$\sin \theta = \frac{b}{a}$$

$$\theta = \arcsin \frac{b}{a}$$

$$A_1 = \frac{a^2}{2} \arcsin \frac{b}{a}$$

Ex: 5.1.3. $\int_0^3 \sqrt{x-1} (x-1) dx$

$$|x| = \sqrt{x^2}$$



$$\int_0^3 \sqrt{\sqrt{(x-1)^2}} (x-1) dx$$

$$\underline{(x-1)^{\frac{3}{4}}}$$

$$\underline{(x-1)^{\frac{3}{2}}}$$

$$\int \sqrt[4]{(x-1)^2} \underline{(x-1)} dx = \int \sqrt[4]{(x-1)^6} dx$$

$$= \int (x-1)^{\frac{6}{4}} dx$$

$$= \left(\frac{1}{\frac{6}{4} + 1} \right) (x-1)^{\frac{6}{4} + 1}$$

$$= \frac{2}{5} \cdot \underline{(x-1)^{\frac{10}{4}}}$$

$$= \frac{2}{5} \cdot (x-1)^{\frac{5}{2}}$$

$$\int_0^3 \sqrt{|x-1|} (x-1) dx = \frac{2}{5} (x-1)^{\frac{5}{2}} \Big|_0^3$$

$$= \frac{2}{5} (4\sqrt{2} - 1)$$

$$\{ (|x|^a x)' = (a+1) |x|^{a-1} x \}$$

$$\underline{\underline{(x)^{\alpha+1})' = (\alpha+1) (x)^{\alpha} x}}$$

$\frac{1}{2} x \text{ co.}$

$$(-x)^{\alpha} x = -(-x)^{\alpha} (-x)$$

$$= -(-x)^{\alpha+1}$$

$$\begin{aligned} [-(-x)^{\alpha+1}]' &= \cancel{(-)} (\alpha+1) (-x)^{\alpha} \\ &= (\alpha+1) (-x)^{\alpha} = (\alpha+1) (x)^{\alpha} \end{aligned}$$

Ex 1:

$$\lim_{n \rightarrow \infty} \left[\left(1 + \frac{1}{n}\right) \left(1 + \frac{2}{n}\right) \cdots \left(1 + \frac{1}{n}\right) \right]^{\frac{1}{n}}$$

$$\begin{aligned} a_n &= \frac{1}{n} \sum_{i=1}^n \ln \left(1 + \frac{i}{n}\right) \\ &= \frac{1}{n} \sum_{i=1}^n \ln \left(1 + \frac{i}{n}\right) \\ \lim_{n \rightarrow \infty} e^{a_n} &= e^{\lim_{n \rightarrow \infty} a_n} = e^{\frac{1}{2} \ln 2} = \frac{e}{4} \end{aligned}$$

$$\boxed{f(x) = \ln(1+x)} \quad \underline{\underline{[0,1] \subset \mathbb{R}}}$$

$$n \sum_{i=1}^n \frac{1}{n}$$

$$\Delta x_i = \frac{1}{n}$$

$$\lim_{n \rightarrow \infty} \sum_{i=1}^n \ln(1 + \xi_i) \Delta x_i = \sum_{i=1}^n \ln(1 + \frac{1}{n}) \frac{1}{n} = \lim_{n \rightarrow \infty} a_n$$

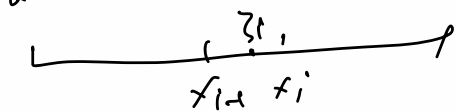
$$\int_0^1 \ln(1+x) dx = \left[(x+1) \ln(1+x) - x \right] \Big|_0^1$$

$$\int \ln(1+x) dx = (x+1) \ln(1+x) - x + C = 2 \ln 2 - 1$$

定理: $f(x)$ 在 $[a, b]$ 上可积,

分割:

$$x_0 < x_1 < x_2 < \dots < x_n$$



$$[x_{i-1}, x_i], i=1, \dots, n$$

$$\Delta x_i = x_i - x_{i-1}$$

$$\xi_i \in (x_{i-1}, x_i)$$

$$\lim_{\lambda \rightarrow 0} \sum_{i=1}^n f(\xi_i) \Delta x_i = I = \int_a^b f(x) dx$$

$$a < b$$

$\forall \varepsilon > 0, \exists \delta > 0$, s.t. 只要 $\lambda < \delta$, 以 λ 为分划 ξ_i 均满足,

$$\left| \sum_{i=1}^n f(\xi_i) \Delta x_i - I \right| < \varepsilon.$$

$$\int_a^a f(x) dx = 0$$

$$\int_b^a f(x) dx = - \int_a^b f(x) dx$$

例: 设 $f(x)$ 在 $[0, 1]$ 上连续, 且 $f(x) > 0$. 证明:

$$\ln \int_0^1 f(x) dx \geq \int_0^1 \ln f(x) dx$$

证明:

$f(x), \ln f(x)$ 在 $[0, 1]$ 上可积

将 $[0, 1]$ 分 n 等份,

$$\Delta x_i = \frac{1}{n}$$

任取 $\xi_i \in [x_{i-1}, x_i]$

e^x

$$\begin{aligned}
 & \lim_{n \rightarrow \infty} e^{\ln \left(\sum_{i=1}^n f(\xi_i) \frac{1}{n} \right)} \geq e^{\sum_{i=1}^n \ln f(\xi_i) \left(\frac{1}{n} \right)} \\
 & = \underline{\underline{\sum_{i=1}^n f(\xi_i) \frac{1}{n}}} \geq = \underline{\underline{\left(f(\xi_1) f(\xi_2) \cdots f(\xi_n) \right)^{\frac{1}{n}}}} \\
 & \quad \text{算术平均} \qquad \qquad \qquad \text{几何平均}
 \end{aligned}$$

定积分的性质.

性质 1. 若 $f(x)$ 在 $[a, b]$ 上可积, k 为一常数, 则 $\underline{\underline{kf(x)}}$ 在 $[a, b]$ 上可积

$$\int_a^b kf(x) dx = k \int_a^b f(x) dx$$

证明: $\lim_{\lambda \rightarrow 0} \sum_{i=1}^n f(\xi_i) \Delta x_i = \int_a^b f(x) dx$

$$\lim_{\lambda \rightarrow 0} \sum_{i=1}^n \left(k f(\xi_i) \right) \Delta x_i = k \lim_{\lambda \rightarrow 0} \sum_{i=1}^n f(\xi_i) \Delta x_i$$

$$\boxed{\sum_{i=1}^n f(\xi_i) \Delta x_i} = \boxed{k \int_a^b f(x) dx}$$

性质2. 若 $f(x)$ 和 $g(x)$ 在 $[a, b]$ 上可积, 则 $f(x) \pm g(x)$ 在 $[a, b]$ 上也可积.

$$\boxed{\int_a^b [f(x) \pm g(x)] dx = \int_a^b f(x) dx \pm \int_a^b g(x) dx}$$

证明.

$$\boxed{\lim_{\lambda \rightarrow 0} \sum_{i=1}^n [f(\xi_i) \pm g(\xi_i)] \Delta x_i}$$

$$= \lim_{\lambda \rightarrow 0} \left\{ \sum_{i=1}^n f(\xi_i) \Delta x_i \pm \sum_{i=1}^n g(\xi_i) \Delta x_i \right\}$$

$$= \lim_{\lambda \rightarrow 0} \sum_{i=1}^n \underbrace{f(\xi_i) \Delta x_i} + \lim_{\lambda \rightarrow 0} \sum_{i=1}^n \underbrace{g(\xi_i) \Delta x_i}$$

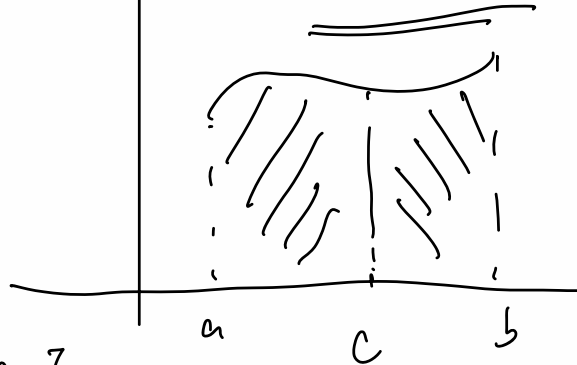
$$= \underbrace{\int_a^b f}_{+} \underbrace{\int_a^b g}$$

性质3. (区间的可加性)

设 $a < c < b$, 若 $f(x)$ 在 $[a, c]$ 和 $[c, b]$ 上可积, 则 $f(x)$ 在 $[a, b]$ 上可积,

$$\underline{\int_a^b f(x) dx} = \boxed{\int_a^c f(x) dx} + \boxed{\int_c^b f(x) dx} \quad (1)$$

$$y = f(x)$$



已知 $f(x)$ 在 I 上可积, $a, b, c \in I$

例: $a < b < c$

$$\int_a^c f(x) dx = \int_a^b f(x) dx + \int_b^c f(x) dx$$

$$\left(\int_a^b f(x) dx \right) = \int_a^c f(x) dx - \int_b^c f(x) dx$$

$$= \left(\int_a^c f(x) dx + \int_c^b f(x) dx \right)$$

特别地. $\int_a^b 1 dx = \int_a^b dx = b - a$

$$\int_a^b c dx = c(b - a)$$

性质 5: (保号性)

设 $f(x)$ 在 $[a, b]$ 上可积, 且 $f(x) \geq 0$, 则 $\int_a^b f(x) dx \geq 0$.

证明:

$$\int_a^b f(x) dx = \lim_{n \rightarrow \infty} \left[\sum_{i=1}^n \frac{1}{2} f(\xi_i) \Delta x_i \right] \geq 0.$$

对任意 ξ_i , $f(\xi_i) \geq 0$, $\Delta x_i > 0$.

推论: (保号性)

在 $[a, b]$ 上 $f(x) \leq g(x)$, 且 $f(x)$ 与 $g(x)$ 均可积.

$$\int_a^b f(x) dx \leq \int_a^b g(x) dx$$

证明:

$$\int_a^b (f(x) - g(x)) dx \leq 0.$$

$$\Leftrightarrow \int_a^b f(x) dx - \int_a^b g(x) dx \leq 0.$$

推论 ②. 若 $f(x)$ 在 $[a, b]$ 上连续且非负, $f(x) \not\equiv 0$.

$$\text{则 } \int_a^b f(x) dx > 0.$$

证明:

存在 $x_0 \in [a, b]$, s.t. $f(x_0) > 0$.

$x_0 \in (a, b)$

$\exists x_0$ 在 $[a, b]$ 内, $\underline{U_\delta(x_0)}$, s.t. $\forall x \in U_\delta(x_0)$,
 $\underline{f(x) \geq \frac{1}{2} f(x_0) > 0}$ $\delta = \frac{1}{2} f(x_0)$

$\exists \cap_{x_0} [\alpha, \beta] \subseteq \underline{U_\delta(x_0)} \subseteq [a, b]$.
 $a < \alpha < \beta < b$.

$$\begin{aligned} \underline{\int_a^b f} &= \underline{\int_a^\alpha f} + \int_\alpha^\beta f + \underline{\int_\beta^b f} \\ &\geq \int_\alpha^\beta f(x) dx \geq \int_\alpha^\beta \underline{\frac{1}{2} f(x_0)} dx \\ &= \underline{\frac{1}{2} f(x_0)} (\underline{\beta - \alpha}) > 0. \end{aligned}$$

推论③: 在 $[a, b]$ 上连续且非负函数 f , $\int_a^b f(x) dx = 0$, 则 $f \equiv 0$.

性质 6. 若 $f(x)$ 在 $[a, b]$ 上可积, 则 $|f(x)|$ 在 $[a, b]$ 上可积, 且

$$\underline{\left| \int_a^b f(x) dx \right|} \leq \int_a^b |f(x)| dx$$

证明:

$$-|f(x)| \leq f(x) \leq |f(x)|$$

$$-\int_a^b |f(x)| dx \leq \int_a^b f(x) dx \leq \int_a^b |f(x)| dx$$

$$\left| \int_a^b f(x) dx \right| \leq \int_a^b |f(x)| dx$$

$$D(x) = \begin{cases} 1 & x \text{ 有理数} \\ 0 & x \text{ 无理数} \end{cases}$$

在 $[0, 1]$ 不可积

$$f(x) = \begin{cases} 1 & x \text{ 有理数} \\ -1 & x \text{ 无理数} \end{cases}$$

$$|f(x)| = 1$$

例: 设 $f(x) = \begin{cases} e^x & x > 0 \\ x^2 & x \leq 0 \end{cases}$

$$\int_{-1}^3 f(x) dx$$

$$\int_{-1}^3 f(x) dx = \int_{-1}^0 f(x) dx + \int_0^3 f(x) dx$$

$$\int_{-1}^0 x^2 dx + \int_0^3 e^x dx$$

$$\begin{aligned}
 &= \int_{-1}^0 \frac{x^3 dx}{e^{2x}} \\
 &= \left. \frac{1}{3} x^3 \right|_{-1}^0 + \left. e^{2x} \right|_{-1}^0 \\
 &= \frac{1}{3} + e^2 - 1 = e^2 - \frac{2}{3}
 \end{aligned}$$

例: $\lim_{n \rightarrow \infty} \int_0^1 \frac{x^n}{1+\sqrt{x}} dx = 0.$ $t=\sqrt{x}$

解:

$$\frac{x^n}{(1+\sqrt{x})} \geq 1 \quad \frac{x^n}{1+\sqrt{x}} \leq x^n$$

$$\begin{aligned}
 0 \leq \int_0^1 \frac{x^n}{1+\sqrt{x}} dx &\leq \int_0^1 x^n dx = \left. \frac{1}{n+1} x^{n+1} \right|_0^1 \\
 &= \frac{1}{n+1} \\
 &\rightarrow 0
 \end{aligned}$$

例: $\lim_{n \rightarrow \infty} \int_0^{\frac{\pi}{2}} \sin^n x dx$

Wallis' formula,

$$\sin x \in [0, \frac{\pi}{2}] \text{ 且 } 1$$

$$\underbrace{0 \leq \sin x \leq 1}_{0 \leq \sin^n x}$$

$$0 \leq x \leq 1$$

$$\sin^n x = 0$$

$$0 \leq \int_0^{\frac{\pi}{2}} \sin^n x \, dx$$

任取 $\varepsilon \in (0, \frac{\pi}{2})$ $(0, \frac{\pi}{2} - \varepsilon)$ $\sin x < 1$
 $(\frac{\pi}{2} - \varepsilon, \frac{\pi}{2})$

$$= \int_0^{\frac{\pi}{2} - \varepsilon} \sin^n x \, dx + \int_{\frac{\pi}{2} - \varepsilon}^{\frac{\pi}{2}} \sin^n x \, dx$$

$$\leq \int_0^{\frac{\pi}{2} - \varepsilon} \sin^n(\frac{\pi}{2} - \varepsilon) \, dx + \int_{\frac{\pi}{2} - \varepsilon}^{\frac{\pi}{2}} 1 \, dx$$

$$= (\frac{\pi}{2} - \varepsilon) \sin^n(\frac{\pi}{2} - \varepsilon) + \varepsilon \leq 2\varepsilon.$$

$$\forall \varepsilon > 0, \exists N, \text{ 当 } n > N \text{ 时 } (\frac{\pi}{2} - \varepsilon) \sin^n(\frac{\pi}{2} - \varepsilon) < \varepsilon.$$

例 5.2.8 (估值定理)

若 $f(x)$ 在 $[a, b]$ 上可积. $m \leq f(x) \leq M$ 则.

$$m(b-a) \leq \int_a^b f(x) \, dx \leq M(b-a)$$

例. $\int_{\frac{\pi}{2}}^{\frac{\pi}{4}} \frac{\sin x}{x} \, dx$ (估值与下界)

$\frac{\pi}{2} > \frac{\pi}{4} \quad x < \tan x$

$$f(x) = \frac{\sin x}{x}$$

$$f'(x) = \frac{x \cos x - \sin x}{x^2} = \frac{\cos x (x - \tan x)}{x^2} < 0.$$

$f(x)$ 在 $[\frac{\pi}{4}, \frac{\pi}{2}]$ 单 ↓

$$f(\frac{\pi}{4}) = \frac{2\sqrt{2}}{\pi}$$

$$f(\frac{\pi}{2}) = \frac{2}{\pi}$$

$$\frac{2}{\pi} \leq f(x) \leq \frac{2\sqrt{2}}{\pi}$$

$$\frac{1}{2} \leq \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} f(x) dx \leq \frac{\sqrt{2}}{2}$$

例 5.2.9 (积分中值定理)

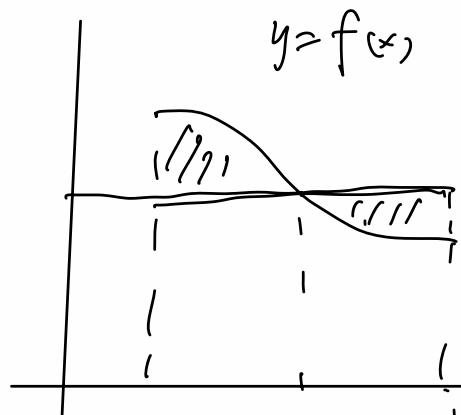
若 $f(x)$ 在 $[a, b]$ 上连续, 则必存在一点 $\xi \in (a, b)$ 使得

$$\int_a^b f(x) dx = f(\xi)(b-a)$$

证:

$\xi \in (a, b)$,

$$f(\xi) = \frac{1}{b-a} \int_a^b f(x) dx$$



$$\underline{m} \leq f(x) \leq \underline{M}$$

$$m(b-a) \leq \frac{\int_a^b f(x) dx}{b-a} \leq M(b-a)$$

||
 μ

介值定理 $\exists \xi \in (a, b)$ s.t. $f(\xi) = \mu = \frac{1}{b-a} \int_a^b f(x) dx$

$$\frac{1}{b-a} \int_a^b f(x) dx$$

$f(x)$ 在 $[a, b]$ 上可积

例 5.2.10 (积分中值定理)

设 $f(x)$ 和 $g(x)$ 在 $[a, b]$ 上连续, $g(x)$ 不改变符号 ($g(x) \geq 0$ 或 $g(x) \leq 0$)

则存在 $\xi \in (a, b)$ 使得

$$\int_a^b f(x)g(x) dx = f(\xi) \int_a^b g(x) dx$$

(当 $g \equiv 1$ 时)

例 5.2.5. 设 $f(x)$ 在 $[0,1]$ 上连续, 在 $(0,1)$ 内可导, 且 $f(1) = 2 \int_0^{\frac{1}{2}} x f(x) dx$

证明: 存在 $\xi \in (0,1)$, s.t. $\boxed{f(\xi) + \xi f'(\xi) = 0}$

证明:

$$(x f(x))' = f(x) + x f'(x)$$

$F(x) = x f(x)$
在 $[0,1]$ 上连续
在 $(0,1)$ 内可导

$\exists \xi \in (0,1)$ s.t. $F'(\xi) = 0$.

$$\begin{aligned} F(0) &= 0 \\ \underline{\underline{F(1)}} &= f(1) = 2 \int_0^{\frac{1}{2}} x f(x) dx \end{aligned}$$

$$= \frac{\int_0^{\frac{1}{2}} F(x) dx}{\underbrace{\left(\frac{1}{2} - 0\right)}} \quad \underline{\underline{\exists c \in (0, \frac{1}{2})}}$$

$$0 < c < \frac{1}{2} < 1$$

$$= \underline{\underline{F(c)}}$$

$F(x)$ 在 $[c,1]$ 上连续

且 $F(c) = F(1) \Rightarrow \exists \xi \in (c,1)$ s.t. $f'(\xi) = 0$.

例: 设 $f(x)$ 可导, 且 $\lim_{x \rightarrow 1} f(x) = 1$.

$$\nexists \lim_{x \rightarrow +\infty} \int_x^{x+2} t \sin \frac{3}{t} f(t) dt$$

$$\nexists \xi \in (x, x+2) \quad \text{s.t.} \quad \int_x^{x+2} t \sin \frac{3}{t} f(t) dt$$

$$= 2 \xi \sin \frac{3}{\xi} f(\xi)$$

$$\frac{1}{2} x \rightarrow +\infty \quad \xi \rightarrow +\infty.$$

$$= \lim_{\xi \rightarrow +\infty} \underbrace{(2)}_{\text{2}} \underbrace{\xi \sin \frac{3}{\xi}}_{\text{3}} \underbrace{f(\xi)}_{\text{f}(\xi)}$$

$$= 2 \lim_{\xi \rightarrow +\infty} \underbrace{\xi \sin \frac{3}{\xi}}_{\text{3}}$$

$$\text{Ab. v.} \quad \frac{\sin \left(\frac{3}{\xi} \right)}{\frac{3}{\xi}} \cdot 3$$

$$= 6$$

例: 5.2.3 (面积引理)

平均

$$\left\{ \int_1^n f(x) dx \right\}$$

设 $f(x)$ 在 $[1, +\infty)$ 上 非负 单减 (Y13.2.42, 证明:

$$\left\{ \sum_{k=1}^n f(k) \right\}$$

$$\frac{\sum_{k=2}^n f(k)}{f(2) + f(3) + \dots + f(n)} < \int_1^n f(x) dx < \frac{\sum_{k=1}^{n-1} f(k)}{f(1) + f(2) + \dots + f(n-1)}$$

证明:

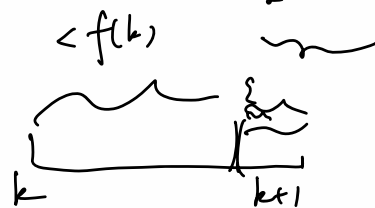
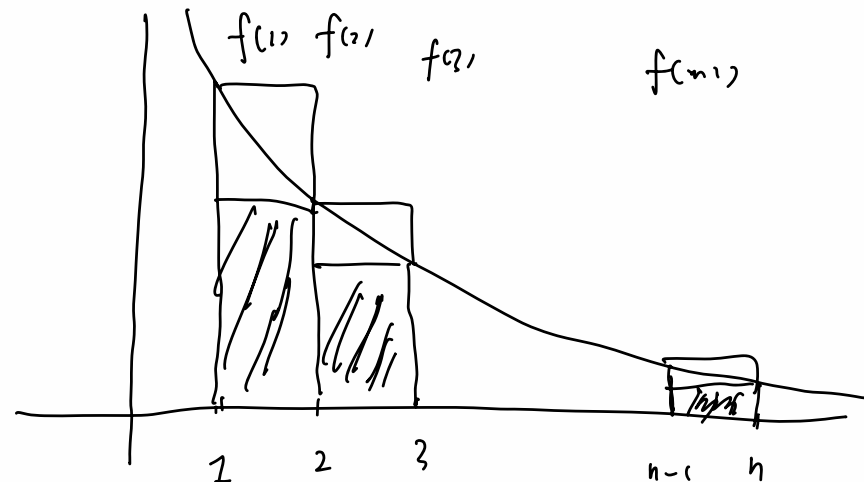
$$\int_1^n f(x) dx = \sum_{k=1}^{n-1} \int_k^{k+1} f(x) dx$$

$f(x)$ 在 $[k, k+1]$ $\downarrow$

$$\underline{f(k+1) \leq f(x) \leq f(k)}$$

$$f(k+1) < \int_k^{k+1} f(x) dx < f(k)$$

$$\underline{\sum_{k=1}^{n-1} f(k+1)} < \int_1^n f(x) dx < \underline{\sum_{k=1}^{n-1} f(k)}$$



$$\sum_{k=2}^n f(k)$$

Young 不等式: 设 $1 < p, q < +\infty$, 且 $\frac{1}{p} + \frac{1}{q} = 1$.

$$\underline{ab \leq \frac{1}{p} a^p + \frac{1}{q} b^q \quad \left(\begin{smallmatrix} \text{特例} \\ \text{柯西} \end{smallmatrix} \quad a^p = b^q \right)}$$

例: S.24 (Hölder 不等式)

设 $f(x), g(x)$ 在 $[a, b] \subset \mathbb{R}$, $1 < p, q < +\infty$, $\frac{1}{p} + \frac{1}{q} = 1$

$$\text{则有} \quad \int_a^b f(x)g(x) dx \leq \left(\int_a^b |f(x)|^p dx \right)^{\frac{1}{p}} \left(\int_a^b |g(x)|^q dx \right)^{\frac{1}{q}}$$

$p = q = 2$. Cauchy - Schwarz

$$\left(\int_a^b \underbrace{f(x)}_{f(x)} \underbrace{g(x)}_{g(x)} dx \right)^2 \leq \int_a^b f^2(x) dx \cdot \int_a^b g^2(x) dx$$

2nd:

$$u(x) = \frac{1}{\left(\int_a^b |f(x)|^p dx\right)^{\frac{1}{p}}}$$

$$u(x) = \frac{1}{\left(\int_a^b |g(x)|^2 dx\right)^{\frac{1}{2}}}$$

$$\int_a^b |u(x)|^p dx = 1$$

$$\int_a^b |v(x)|^p dx = 1$$

$$\underline{\underline{|u(x)| |v(x)| \leq \frac{1}{p} |u(x)|^p + \frac{1}{q} |v(x)|^q}}$$

作业:

① 证明定理 5.2.10

② 设 $f(x)$ 是 $[0,1]$ 上的连续函数, 且 $f > 0$, 证明:

$$\int_0^1 f(x) dx \int_0^1 \frac{1}{f(x)} dx \geq 1$$

(不要用 Hölder 不等式)

§5.3 微积分基本定理.

设 $f(t)$ 在 $[a, b]$ 上可积, $\forall x \in [a, b]$, $f(t)$ 在 $[a, x]$ 上也可积.

$$\underline{\underline{\int_a^x f(t) dt}}$$

定义 5.3.1 $f(t)$ 在 $[a, b]$ 上可积, $\int_a^x f(t) dt$ 称为 $f(t)$ 在 $[a, b]$ 上的变上限积分.

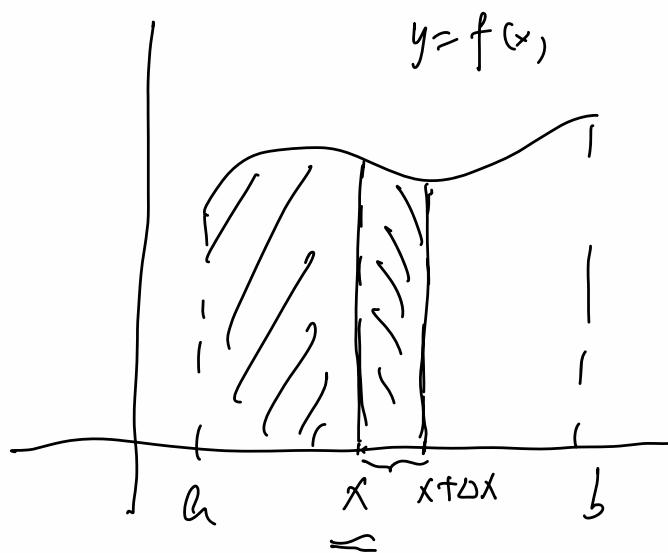
$$F(x) = \int_a^x f(t) dt, \quad x \in [a, b]$$

$$F(a) = \int_a^a f(t) dt = 0.$$

$$F(b) = \int_a^b f(t) dt$$

$$\int_x^b f(t) dt$$

$$A = \int_x^{x+\Delta x} f(t) dt \leq \underline{\underline{M \cdot \Delta x}}$$



定理 5.3.1. 设 $f(x)$ 在 $[a, b]$ 上可积, 那么 $F(x) = \int_a^x f(t) dt$ 在 $[a, b]$ 上连续.

证明:

$\left\{ \begin{array}{l} (a, b) \text{ 连续} \\ a \text{ 处右连续} \end{array} \right.$

b 处左连续

$$\forall \varepsilon > 0, \exists \delta = \frac{\varepsilon}{M}.$$

s.t. $|x - x_0| < \delta$

$$|F(x) - F(x_0)| < M \cdot \delta = \varepsilon.$$

任取 $x_0 \in (a, b)$, 充分小的 h

$$\begin{aligned} \underline{F(x_0+h) - F(x_0)} &= \int_a^{x_0+h} f(t) dt - \int_a^{x_0} f(t) dt \\ &= \int_a^{x_0+h} f(t) dt + \int_{x_0}^a f(t) dt \\ &= \underline{\int_{x_0}^{x_0+h} f(t) dt} \end{aligned}$$

令 M 是 $f(x)$ 的上界

$$|f(x)| \leq M.$$

$$|F(x_0+h) - F(x_0)| \leq \underline{\int_{x_0}^{x_0+h} |f(t)| dt} \leq M \cdot h.$$

定理 5.3.2. 设 $f(x)$ 在 $[a, b]$ 上可积, 在 $x_0 \in [a, b]$ 处是连续的. 那么

$$F(x) = \int_a^x f(t) dt \text{ 在 } x_0 \text{ 处可导, 且 } F'(x_0) = f(x_0).$$

证明:

$$F(x_0+h) - F(x_0) = \int_{x_0}^{x_0+h} f(t) dt$$

$$\lim_{h \rightarrow 0} \frac{F(x_0+h) - F(x_0)}{h} = f(x_0)$$

$$\left| \frac{F(x_0+h) - F(x_0)}{h} - f(x_0) \right| < \varepsilon.$$

$f(x)$ 在 x_0 处连续 $\Rightarrow \forall \varepsilon > 0, \exists \delta > 0$, s.t. $\underline{|t - x_0| < \delta}$ 且 $t \in [a, b]$, 都有

$$|f(t) - f(x_0)| < \varepsilon$$

$$\Leftrightarrow \underline{f(x_0) - \varepsilon < f(t) < f(x_0) + \varepsilon}$$

$$\text{且 } \forall x = |h| < \delta, \quad \underline{(x_0, x_0+h)} \subseteq (x_0 - \delta, x_0 + \delta)$$

$$(f(x_0) - \varepsilon)h < \int_{x_0}^{x_0+h} f(t) dt < (f(x_0) + \varepsilon)h$$

$$\underline{f(x_0) - \varepsilon < \frac{1}{h} \int_{x_0}^{x_0+h} f(t) dt < f(x_0) + \varepsilon}$$

$$\underline{\left| \frac{1}{h} F(x_0+h) - F(x_0) - \underline{f(x_0)} \right| < \varepsilon.}$$

定理 5.3.3 (微积分基本定理)

设 $f(x)$ 在 $[a, b]$ 上连续, $c \in [a, b]$, 则

$$\frac{d}{dx} \int_c^x f(t) dt = f(x), \quad x \in [a, b].$$

$$\begin{aligned} \int_c^x f(t) dt &= \int_c^a f(t) dt + \int_a^x f(t) dt \\ &= \int_a^x f(t) dt - \int_a^c f(t) dt \end{aligned}$$

设 $F(x)$ 是 $f(x)$ 在 $[a, b]$ 上的一个原函数

$$F(x) = \int_a^x f(t) dt + C \Rightarrow \int_a^x f(t) dt = \frac{F(x) - F(a)}{x \in [a, b]}$$

当 $x = a$ 时 $F(a) = C$

定理 5.3.4 若 $F(x)$ 在 $[a, b]$ 上可导且导函数连续,

$$\int_a^x F'(t) dt = F(x) - F(a), \quad x \in [a, b].$$

可以

由 $F'(t) = f(t)$ 得 $\int_a^x f(t) dt = F(x) - F(a)$

①. $f(x) = \begin{cases} 1 & x > 0 \\ 0 & x = 0 \\ -1 & x < 0 \end{cases}$

②. 积分 - 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83 84 85 86 87 88 89 90 91 92 93 94 95 96 97 98 99 100

例. $f(x) = \text{sgn} x = \begin{cases} 1 & x > 0 \\ 0 & x = 0 \\ -1 & x < 0 \end{cases} \rightarrow \text{第一类间断点.}$

$[-b, b]$ 上可积.

例. $f(x) = \begin{cases} 2x \sin \frac{1}{x^2} - \frac{2}{x} \cos \frac{1}{x^2}, & x \neq 0 \\ 0, & x = 0 \end{cases}$

在 $[-b, b]$ 上可积.

$F(x) = \begin{cases} x^2 \sin \frac{1}{x^2}, & x \neq 0 \\ 0, & x = 0 \end{cases}$

例. 5.3.1. 若 $f(x), g(x)$ 在 $[a, b]$ 上连续, 且 $\int_a^x f(t) dt = g(x)$, $\Leftrightarrow f(x) = g'(x)$, 且 $g(a) = 0$.

证明: 左 $\Rightarrow$ 右 设 $x = a \Rightarrow g(a) = 0$.

$$f(x) = g'(x)$$

$$t_1 \Rightarrow t_2 \quad f(x) = g'(x)$$

$$\int_a^x f(t) dt = \int_a^x g'(t) dt = \underline{\underline{g(x) - g(a)}} \\ \underline{\underline{= g(x).}}$$

例: $F(x) = \int_{x^2}^0 \sqrt{1+t^4} dt$, $\frac{d}{dx} F(x)$.

解: $\underline{\underline{F(x) = - \int_0^{x^2} \sqrt{1+t^4} dt}}$

$$\underline{\underline{u = x^2}}$$

$$\underline{\underline{F(u) = - \int_0^u \sqrt{1+t^4} dt}}$$

for $u = x^2$ it's $\frac{2}{2}$.

$$\underline{\underline{F(x) = F \circ u = F(u(x))}}$$

$$F'(x) = \frac{dF}{du} \cdot \frac{du}{dx}$$

$$= -\sqrt{1+u^4} \cdot (2x) = -2x \cdot \sqrt{1+x^8}$$

例 5.3.5.

若 $f(x)$ 在 (a, b) 内连续, $\alpha(x)$ 和 $\beta(x)$ 可导, 且 $\alpha(x)$ 和 $\beta(x) \in (a, b)$.

$$\frac{d}{dx} \int_{\alpha(x)}^{\beta(x)} f(t) dt = f(\beta(x)) \beta'(x) - f(\alpha(x)) \alpha'(x).$$

例 5.3.4. 已知 $\lim_{x \rightarrow 0} \frac{1}{bx - \sin x} \left[\int_0^x \frac{t^2}{\sqrt{a+t^2}} dt \right] = I$, $\forall a, b > 0$ 求 I .

解: $\frac{0}{0}$ 型

$$\downarrow$$

$$= \lim_{x \rightarrow 0} \frac{\frac{x^2}{\sqrt{a+x^2}}}{b - \cos x} = I.$$

$$\lim_{x \rightarrow 0} (b - \cos x) = 0 \Rightarrow b = 1 \uparrow$$

$$= \lim_{x \rightarrow 0} \frac{\frac{x^2}{\sqrt{a+x^2}}}{1 - \cos x} = \lim_{x \rightarrow 0} \left(\frac{1}{\sqrt{a+x^2}} \right) \cdot \frac{x^2}{1 - \cos x}$$

$$= 1 \cdot \lim_{x \rightarrow 0} \frac{x^2}{1 - \cos x} = \frac{2}{1} = 2$$

$$\lim_{x \rightarrow 0} \frac{x}{1 - \cos x} = \frac{0}{0} = 1$$

$$1 - \cos x \sim \frac{1}{2}x^2 \Rightarrow a=4$$

例: $\lim_{x \rightarrow 0} \frac{\int_{\cos x}^1 e^{-t^2} dt}{x^2}$

$$= - \lim_{x \rightarrow 0} \frac{\int_1^{\cos x} \underbrace{e^{-t^2}}_{dt} dt}{x^2} = - \lim_{x \rightarrow 0} \frac{e^{-\cos^2 x} (-\sin x)}{2x}$$

$$= \lim_{x \rightarrow 0} \frac{1}{2} \cdot \underbrace{\frac{\sin x}{x}}_1 \cdot \underbrace{e^{-\cos^2 x}}_1 = \frac{1}{2}$$

$$= \frac{1}{2e}$$

例: 设 $f(x)$ 在 $(-a, a)$ 内连续, 且 $f(x) > 0$.

证明 $F(x) = \frac{\int_0^x t f(t) dt}{\int_0^x f(t) dt}$

在 $(0, a)$ 内单调递增.

证明: $\frac{d}{dx} \int_0^x t f(t) dt = x f(x)$